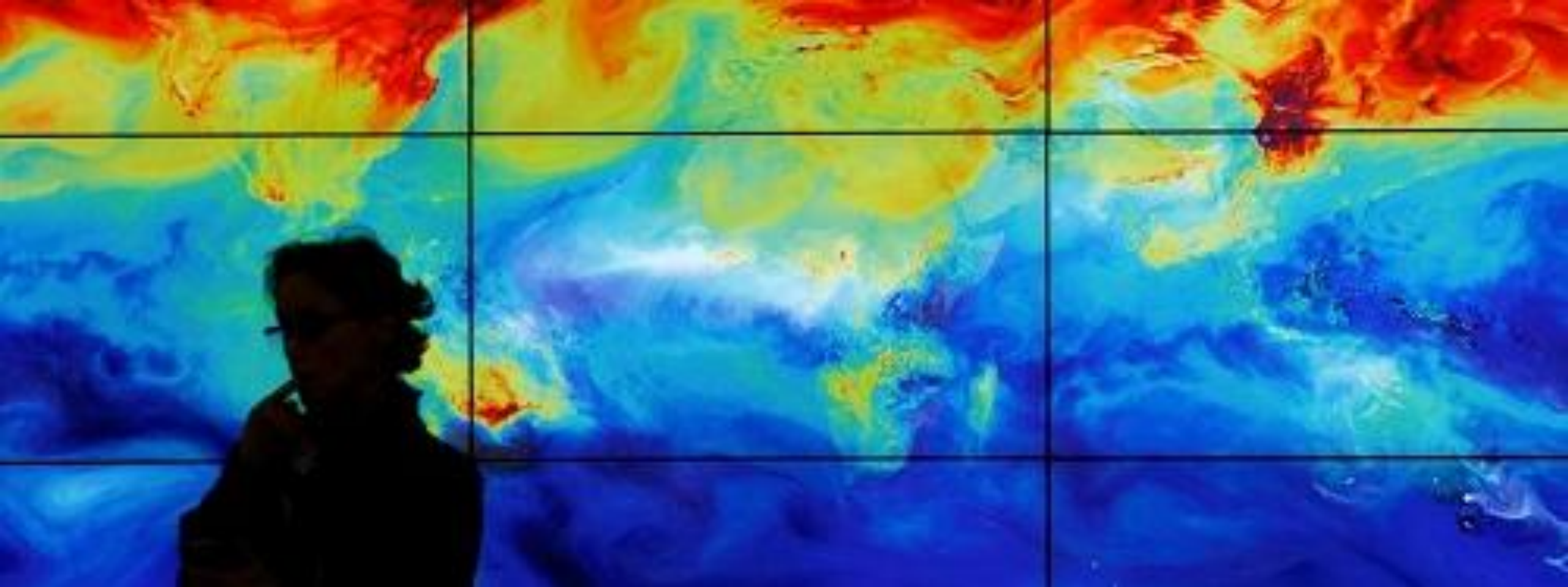




Climate Clustering of European Regions

A Comprehensive Novel Study on Long-Term Hazard Forecast, Risk Exposure, and Vulnerability to Incidents at the NUTS3 Level.

Agenda

- Project Objective
- Methodological Framework
- Benchmark Clustering Results
- Cluster Explainability & Interpretability
- Scenario Analysis: Flooding Case Study
- Future Research Directions



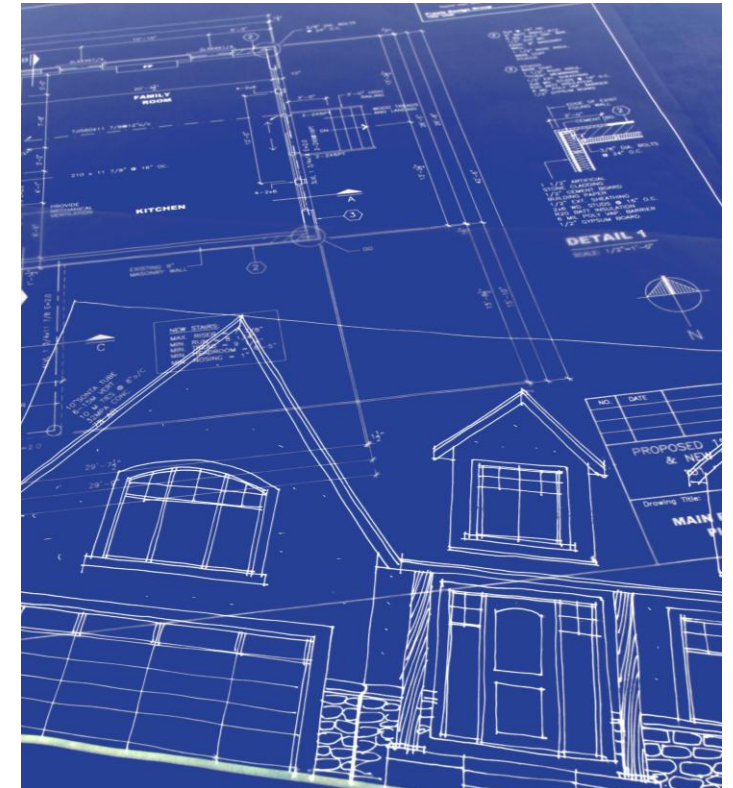
Project Objective

- To identify clusters of similar European regions based on **environmental, socio-economic and infrastructure indicators**, aiming to uncover patterns of climate risk and resilience.
- The goal is to develop a **data-driven typology** that helps interpret **regional vulnerabilities and adaptation capacities** across Europe.



Methodology

- Collected and harmonized data aligned (as closely as possible) with the RESIN Project framework (hazard, exposure, and vulnerability indicators).
- Identified the optimal combination of algorithms that best replicate regional clustering patterns.
- Determined the key attributes driving the formation of clusters using interpretability techniques.
- Leveraged the methodology to enable scenario-based clustering for specific hazards. e.g., *Flooding, Wildfires, Cold spills* etc.



About RESIN Project

- RESIN Project created by EUROPA, a climate-risk typology using expert knowledge.
- Regions were grouped by hazard, exposure, and socio-economic resilience.
- Defined distinct regional profiles (e.g., Lowlands, Urbanized, Southern Lands).

North west coasts

C: coastal
KH: coastal flooding
E: high (high exposure of population and infrastructure to coastal hazards)
S: average
AC: average

Lowlands and estuaries

C: coastal
KH: coastal and river flooding, heavy precipitation
E: very high exposure of population and infrastructure to coastal hazards and river flooding
S: average (relatively low proportion of population at risk of poverty)
AC: relatively high (relatively high GDP, good infrastructure provision, higher than average innovation levels)

North west urban

C: mainly inland cities
KH: heavy precipitation
E: lower than average exposure of population and infrastructure to fluvial flooding and landslides
S: average
AC: relatively high (high transport infrastructure density, higher than average GDP, relatively high innovation levels)

Inland and urbanised

C: inland, large cities, including capital cities
KH: river flooding, heavy precipitation
E: relatively high exposure of population and transport infrastructure to river flooding
S: low (projected increase of young people)
AC: high (affluent cities, high provision of critical infrastructure, high innovation levels)

Northern lands

C: low density, green cities and small rural settlements
KH: coastal hazards; heavy precipitation; low and high temperatures
E: higher than average exposure of population and infrastructure to coastal flooding
S: low (projected increase of young people and immigrants)
AC: high (high provision of critical infrastructure, high innovation levels)

Inland hinterlands

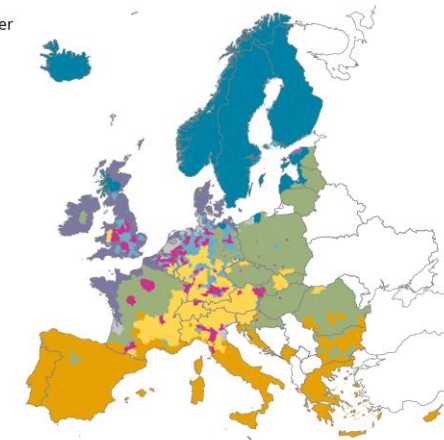
C: peri-urban and rural settings, relatively low population densities
KH: multiple climate hazards
E: relatively high exposure of population and transport infrastructure to river flooding
S: average, with projected decrease in young people
AC: lower than average (lower than average GDP and employment opportunities; high priority allocation funding; low innovation levels)

Landlocked and elevated

C: mountainous and upland areas, relatively low population densities
KH: landslides, river flooding, heavy precipitation
E: high exposure of population and infrastructure to landslides
S: lower than average (lower than average proportion of people at risk of poverty)
AC: higher than average (relatively affluent areas, dense road infrastructure; relatively high innovation levels)

Southern lands

C: Mediterranean, high density
KH: high temperatures, droughts, wildfires, landslides, coastal hazards
E: higher than average exposure of transport infrastructure and population to landslides
S: relatively high (higher than average population at risk of poverty)
AC: relatively low (lower than average GDP, relatively low critical infrastructure provision, lower than average innovation levels)



Our study aims to reproduce and extend this typology algorithmically. Which enables objective, data-driven and scenario-based regional clustering.

Data Overview of the RESIN Project

☐ Hazard

- Temperature
- precipitation
- flooding
- drought
- landslide

☐ Exposure

- population
- roads
- railways
- transport nodes exposed to hazards

☐ Vulnerability

- ageing population
- employment balance
- infrastructure density

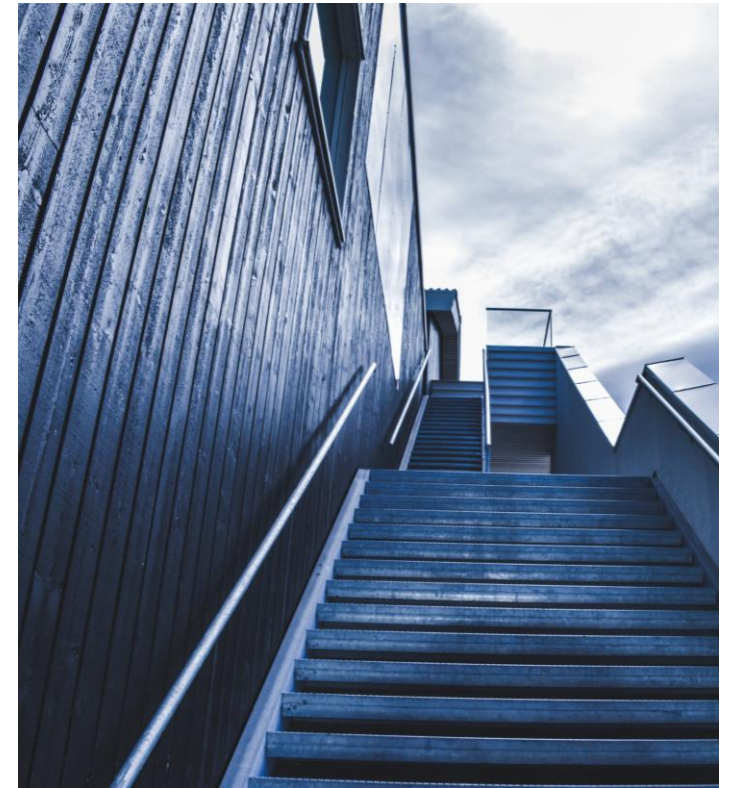
Data overview for Climate Clustering

- Reconstructed 30+ indicators out of ~45 defined in RESIN.
Some secondary indicators (e.g., broadband, hospitals per head, patents) were unavailable due to discontinued EU datasets.
- Non-trivial data processing effort: required multi-source acquisition, cleaning, geospatial aggregation, and standardization across Europe.
- Infrastructure variables (roads, rail, transport nodes) are key as they heavily influence exposure and vulnerability scores.

Hazard Indicators	Exposure Indicators	Vulnerability Indicators
Mean Temperature	Population in Settlements Exposed to Fluvial Flooding	Migratory Population Change
Maximum Temperature	Population in Settlements Exposed to Coastal Hazards	Population Change – Children
Summer Days	Population in Settlements Exposed to Landslides	Population Change – Older People
Heat Waves	Road Infrastructure Exposed to Fluvial Flooding	Employment–Population Balance
Ice Days	Rail Network Exposed to Fluvial Flooding	Length of Major Road Networks
Consecutive Wet Days	Road Infrastructure Exposed to Coastal Hazards	Length of Railway Network
Heavy Precipitation	Rail Network Exposed to Coastal Hazards	Density of Major Road Intersections
Very Heavy Precipitation	Road Infrastructure Exposed to Landslides	Density of Transport Nodes
Consecutive Dry Days	Rail Network Exposed to Landslides	Airports per Head of Population
Wildfire	—	Ports per Head of Population
Coastal Hazard	—	Urban Area Classified as Green Space
Drought Hazard	—	Change in Total Green Space
Fluvial Hazard	—	Urban Land Cover
Landslide Hazard	—	Change in Land Cover

Key Steps

- Converted all geospatial data in the same format
- Integrated and harmonized datasets
- Reduced Dimensionality of the data
- Applied different clustering techniques
- Compared outcome of clustering to identify best methodology
- Interpreted underlying indicators per cluster
- Applied the best discovered technique for scenario analysis

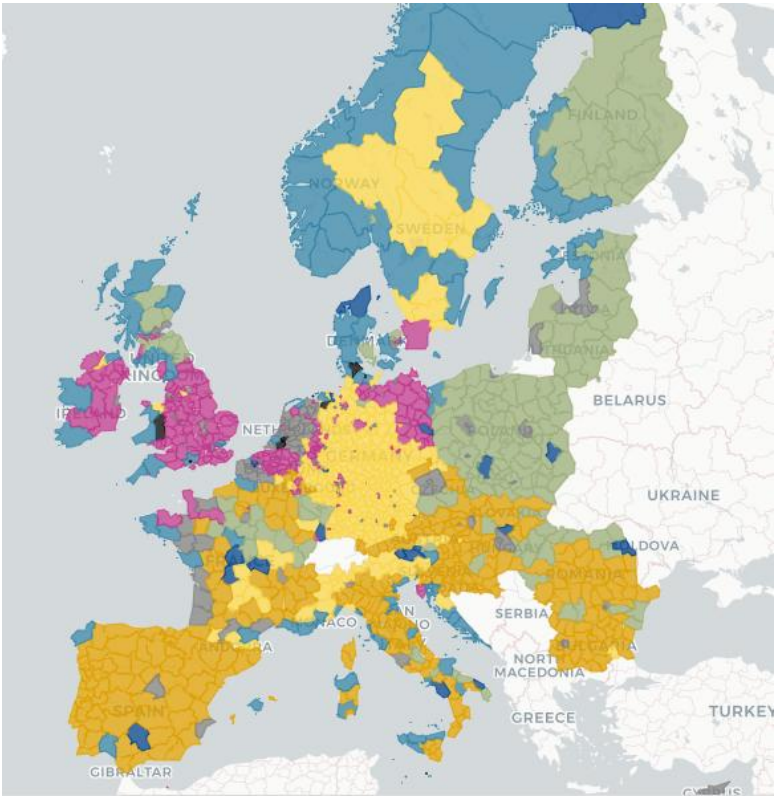


Results: Internal Vs External Validation

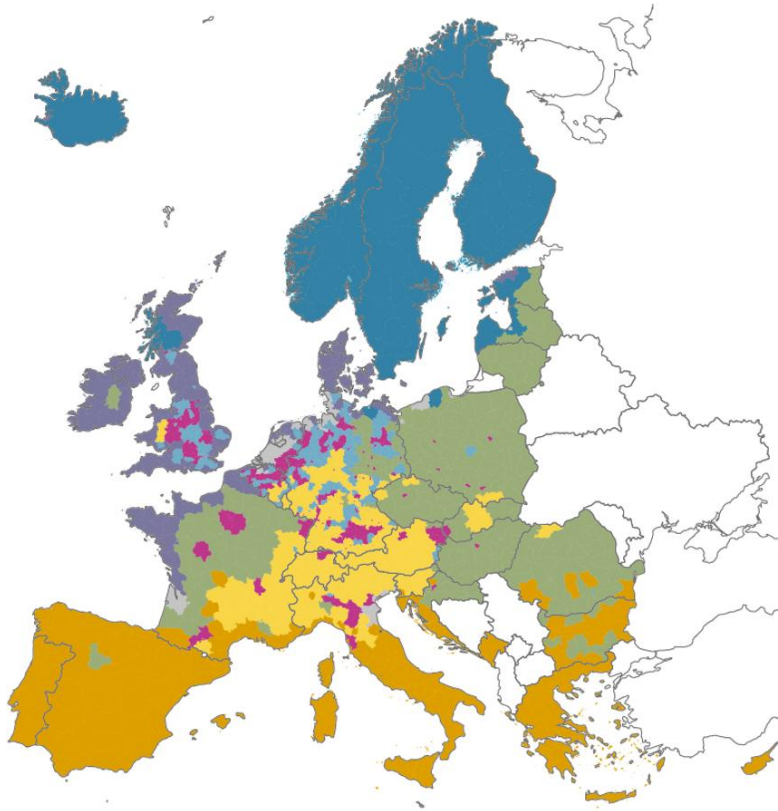
- 1
- 2
- 3

Rank ⌘	Reduction Technique	Clustering Algorithm	Number of Clusters (k)	External Validation			Internal Validation		
				Adjusted Rand Index (ARI)	Rand Index	Normalized Mutual Information (NMI)	Silhouette Score	Davies–Bouldin Index	Calinski–Harabasz Index
1	KPCA (RBF $\gamma=0.001480$)	K-Means	8	0.057	0.690	0.241	0.362	1.155	323
2	KPCA (RBF $\gamma=0.001480$)	Spectral (Nearest Neighbors)	10	0.188	0.870	0.319	0.193	1.391	664
3	KPCA (RBF $\gamma=0.001480$)	Agglomerative (Average Linkage)	8	0.025	0.300	0.089	0.490	0.557	623
4	KPCA (RBF $\gamma=0.001480$)	GMM (Full Covariance)	8	0.149	0.720	0.235	0.226	1.352	821
5	KPCA (RBF $\gamma=0.044398$)	Spectral (Nearest Neighbors)	8	0.177	0.753	0.315	0.209	1.486	205
6	KPCA (RBF $\gamma=0.044398$)	Spectral (Nearest Neighbors)	10	0.161	0.741	0.309	0.200	1.447	202
7	KPCA (RBF $\gamma=0.001480$)	GMM (Full Covariance)	10	0.155	0.705	0.259	0.188	1.387	743
8	KPCA (RBF $\gamma=0.001480$)	GMM (Diagonal Covariance)	10	0.155	0.705	0.259	0.143	1.515	571
9	UMAP (10 Neighbors)	K-Means	7	0.122	0.672	0.201	0.457	0.766	1076
10	UMAP (10 Neighbors)	K-Means	6	0.120	0.670	0.199	0.457	0.793	1062
11	UMAP (10 Neighbors)	Spectral (Nearest Neighbors)	7	0.117	0.665	0.195	0.443	0.772	992
12	KPCA (RBF $\gamma=0.001480$)	Agglomerative (Ward)	10	0.142	0.710	0.244	0.212	1.374	809
13	KPCA (RBF $\gamma=0.001480$)	K-Means	10	0.138	0.703	0.239	0.204	1.388	777
14	KPCA (RBF $\gamma=0.044398$)	Agglomerative (Average Linkage)	8	0.151	0.725	0.260	0.198	1.401	798
15	KPCA (RBF $\gamma=0.044398$)	K-Means	8	0.147	0.719	0.258	0.189	1.412	786

Side-by-Side Comparison: Our Method Vs RESIN's Expert Labels

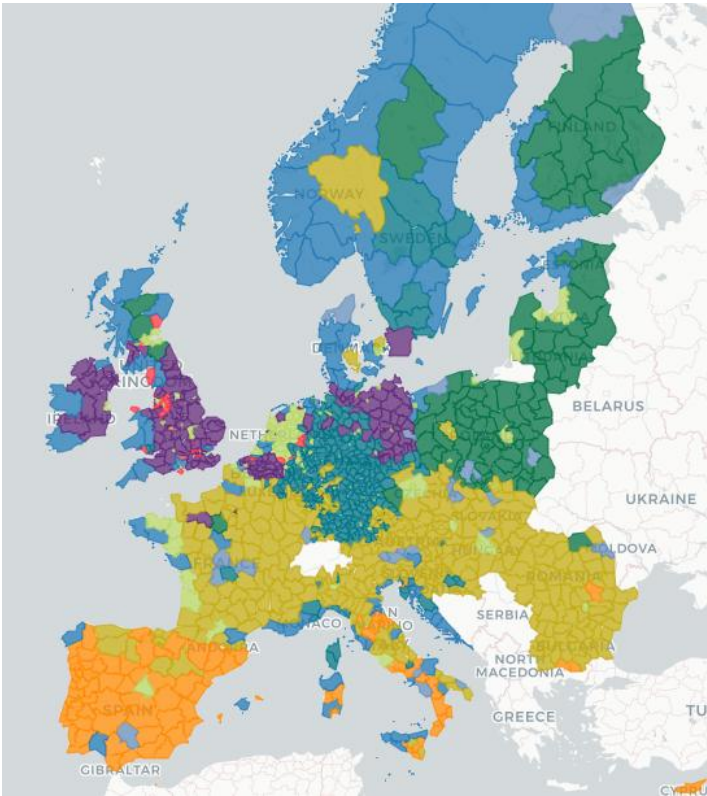


K means

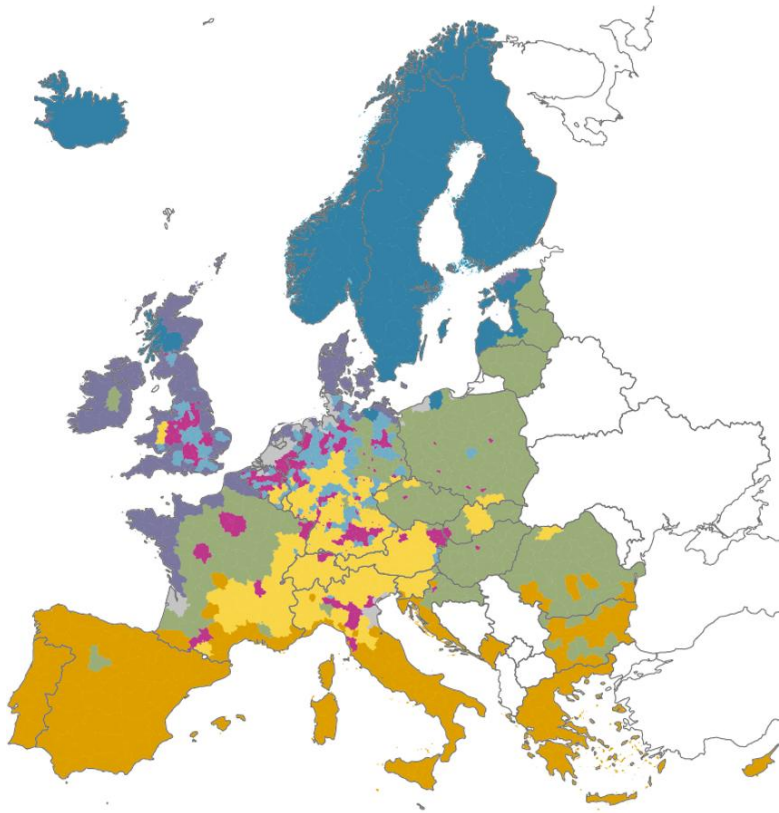


RESIN

Side-by-Side Comparison: Our Method Vs RESIN's Expert Labels



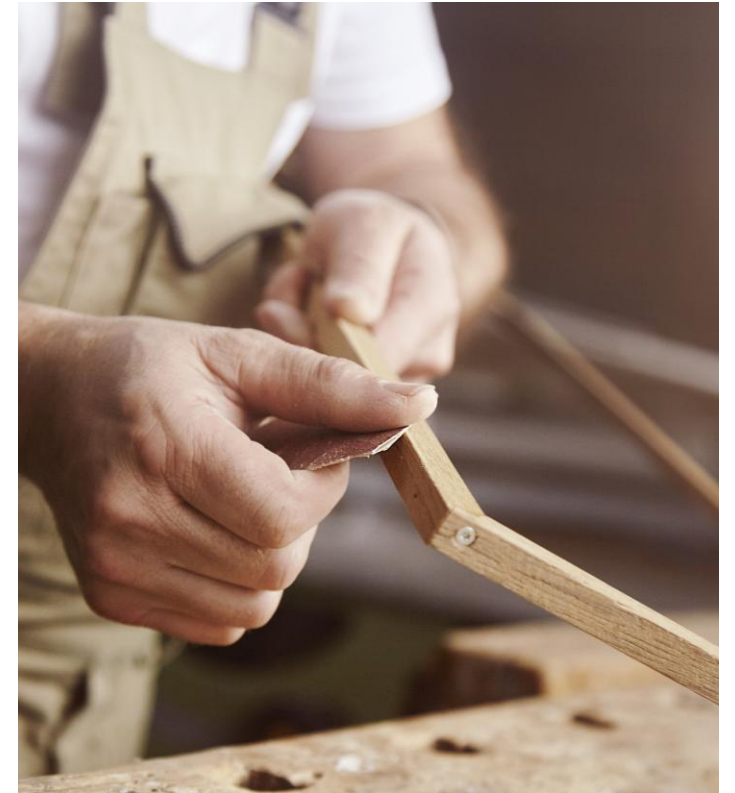
Spectral_nn



RESIN

Conclusion So far...

- After testing hundreds of algorithmic combinations, this configuration provides the best match to the original RESIN typology.
- This proves that expert-based classifications can be reproduced algorithmically with strong consistency.

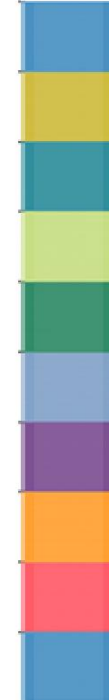
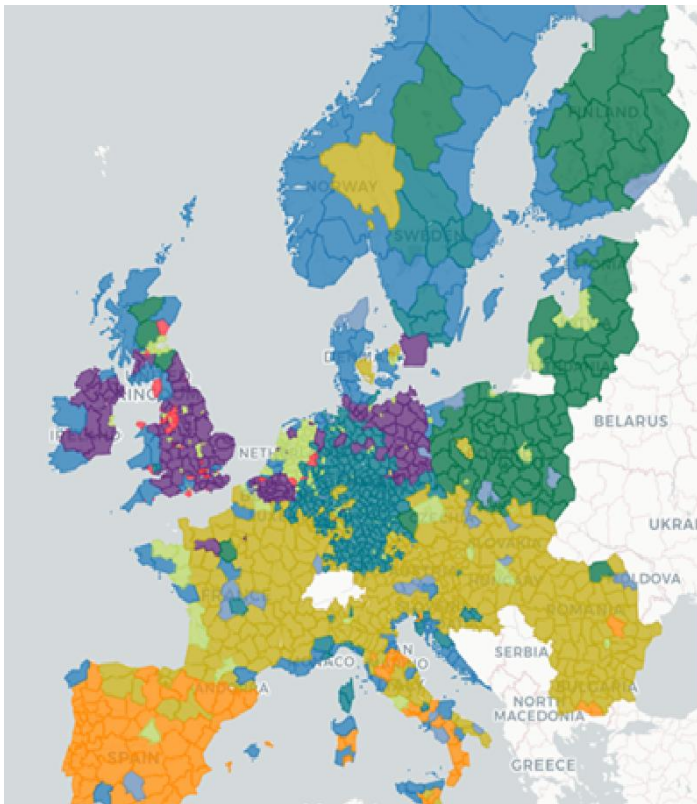


Why We Need Explainability ?

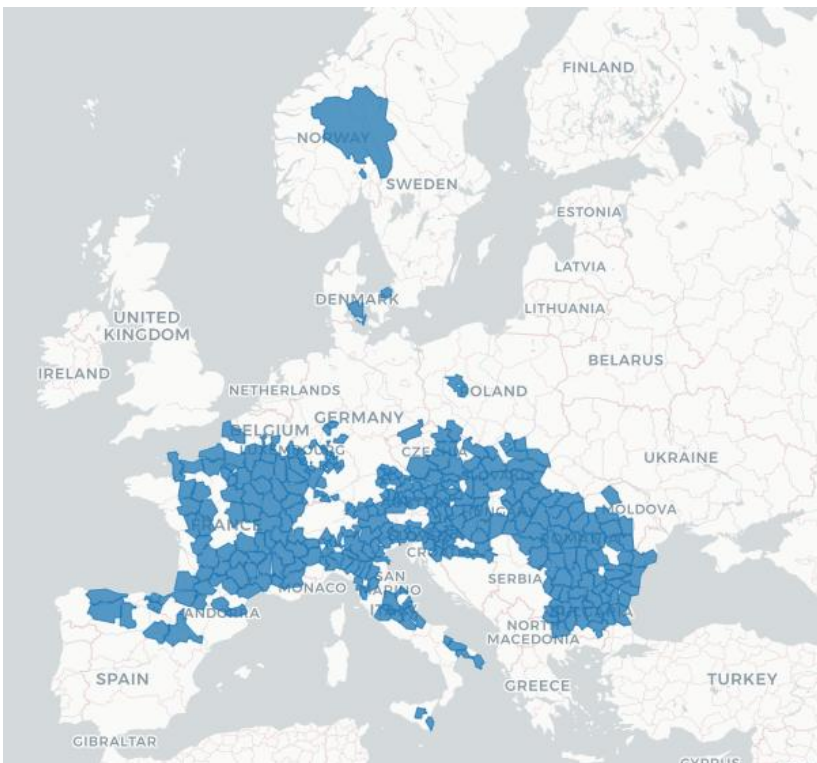
- Algorithms act as **black boxes**, they reveal groups but not why those groups exist.
- **Understand which indicators drive each cluster**, making results interpretable and actionable.
- By finding underlying features, determine how each cluster is different than others.
- This approach **closes the loop**, turning abstract algorithmic outputs into clear insights.



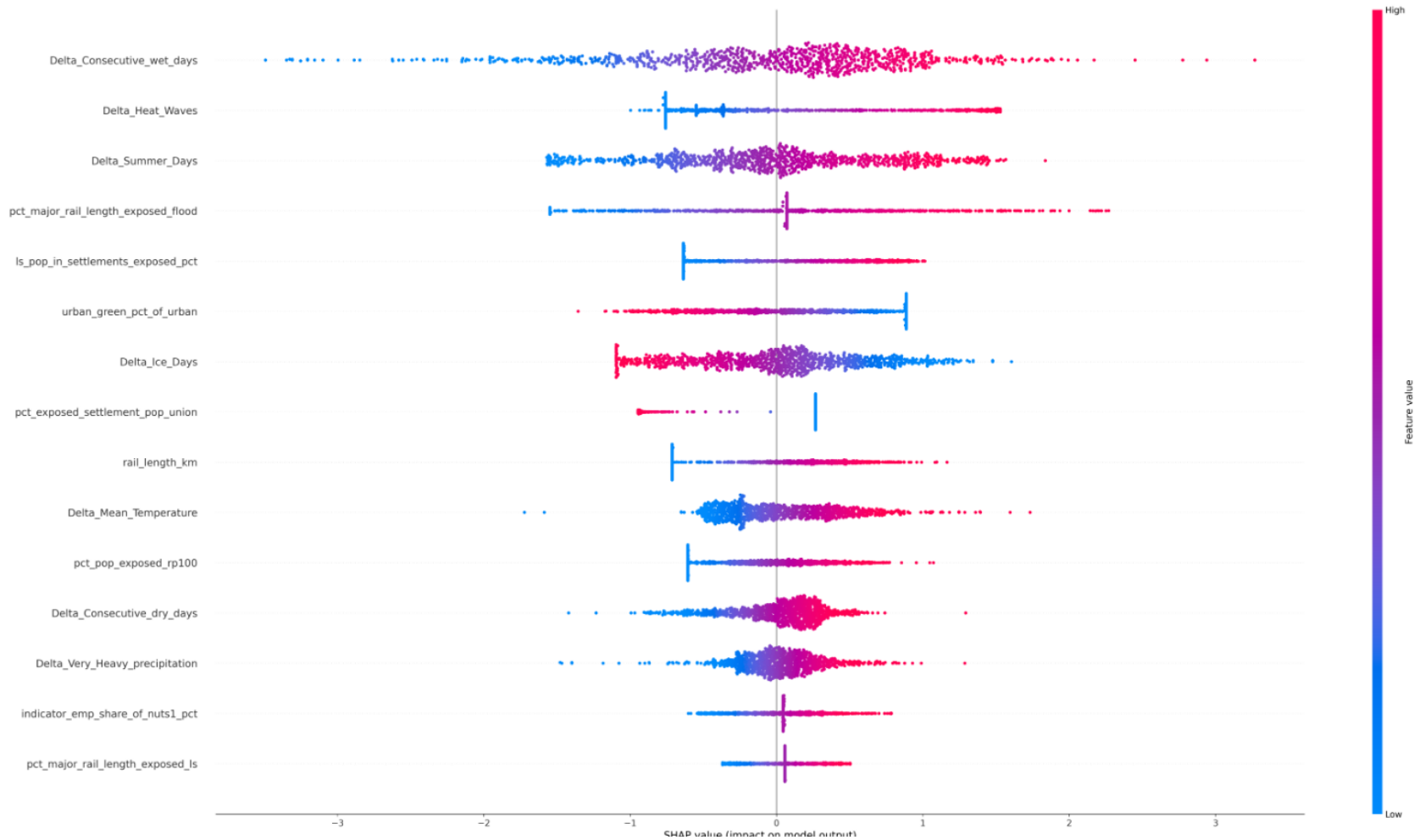
Spectral_nn

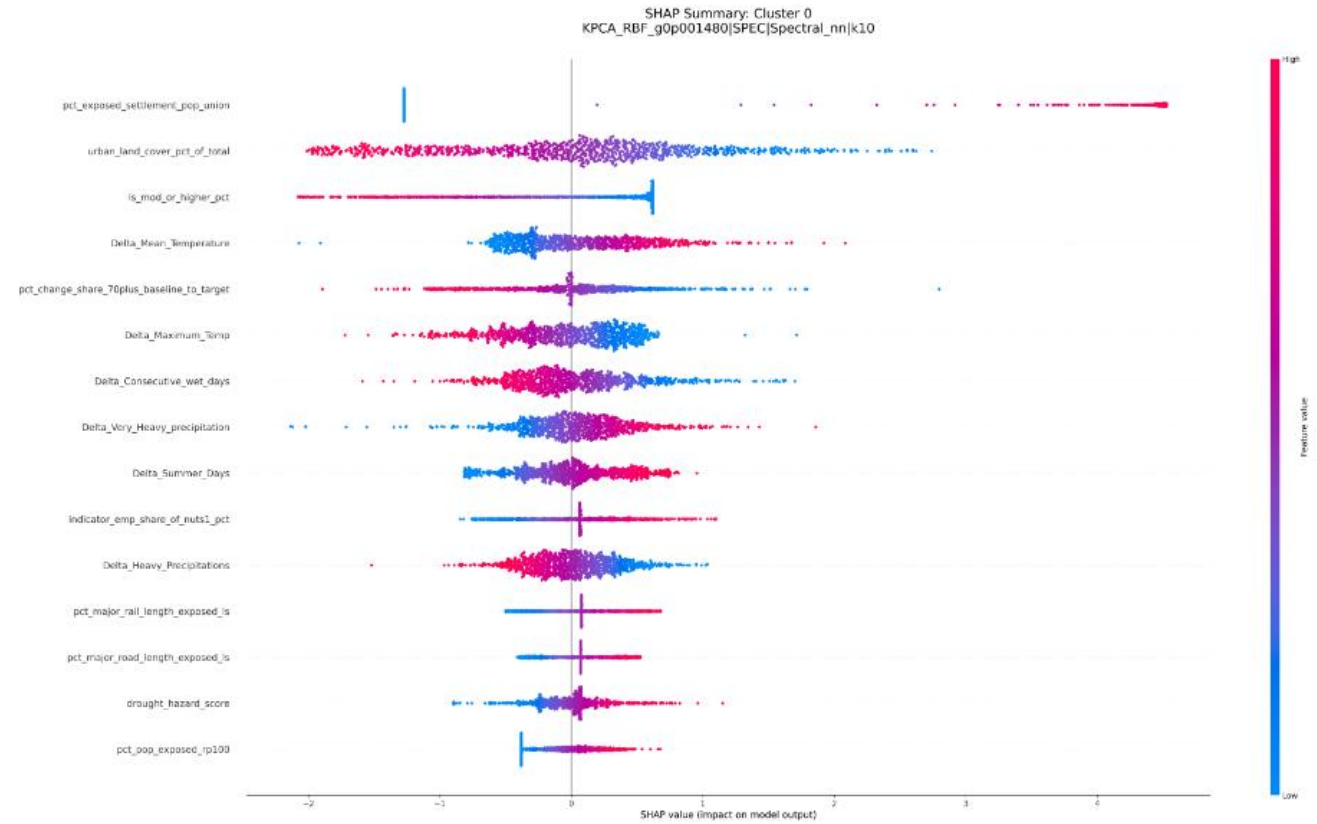
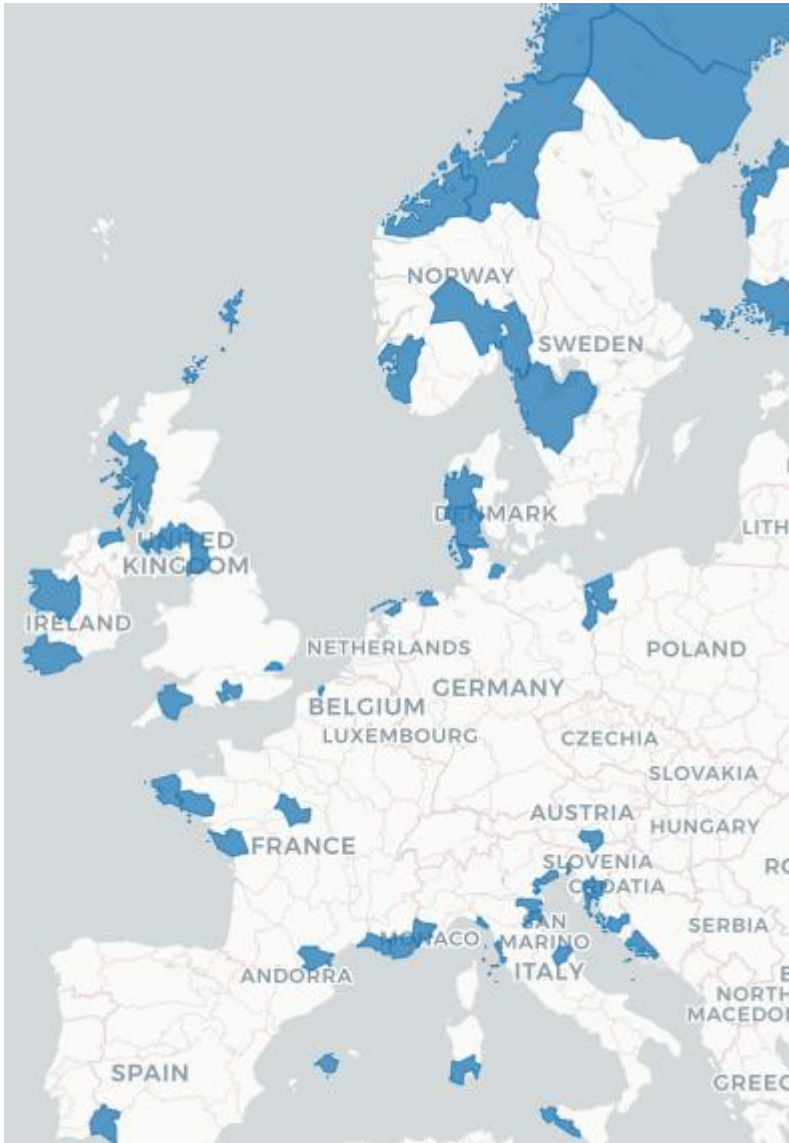


Cluster	Dominant Drivers (Top SHAP Features)	Theme	Typology Label
0	Low Ports Density, High Green Share, Stable Climate Conditions	Low Hazard & High Adaptation	Stable Humid Transition Zones
1	High Flood Exposure, Rapid Land-Cover Change, Economic Activity	Hydro-Climate Driven Urban Growth	Urban Floodplain Expansion Zones
2	High Heat Wave Frequency, Long Dry Periods, Low Green Cover	Thermal Hazard Dominance	Urban Heat-Stressed Regions
3	High Urban Land Cover, Heavy Precipitation, Transport Density	Urban–Climate Pressure	Metropolitan Areas Under Climate Stress
4	Ageing Population, Low Exposure, Decreasing Green Urban Share	Demographic Vulnerability + Limited Adaptation	Aging Low-Exposure Hinterlands
5	High Migration Status	Medium Hazard Exposure	Migration Zone with Rail Connectivity
6	Low Exposure, High Employment, High Education Share	Socio-Economic Strength, Low Hazard	Resilient Northern Urban Centers
7	Heat Exposure, Future Drought Zone, High Economic Activity	Transport & Heat Exposure Corridors	Industrial and Heat Climate Corridors
8	Low Exposure, High Drought Index, Fewer Wet Days	Dry Transitional Zones	Arid Southern Transition Regions
9	High Settlement Exposure, Increasing Heavy Precipitation, Moderate Urban Green Share	Fluvial Risk with Moderate Adaptation	Expanding Riverine Urban Settlements



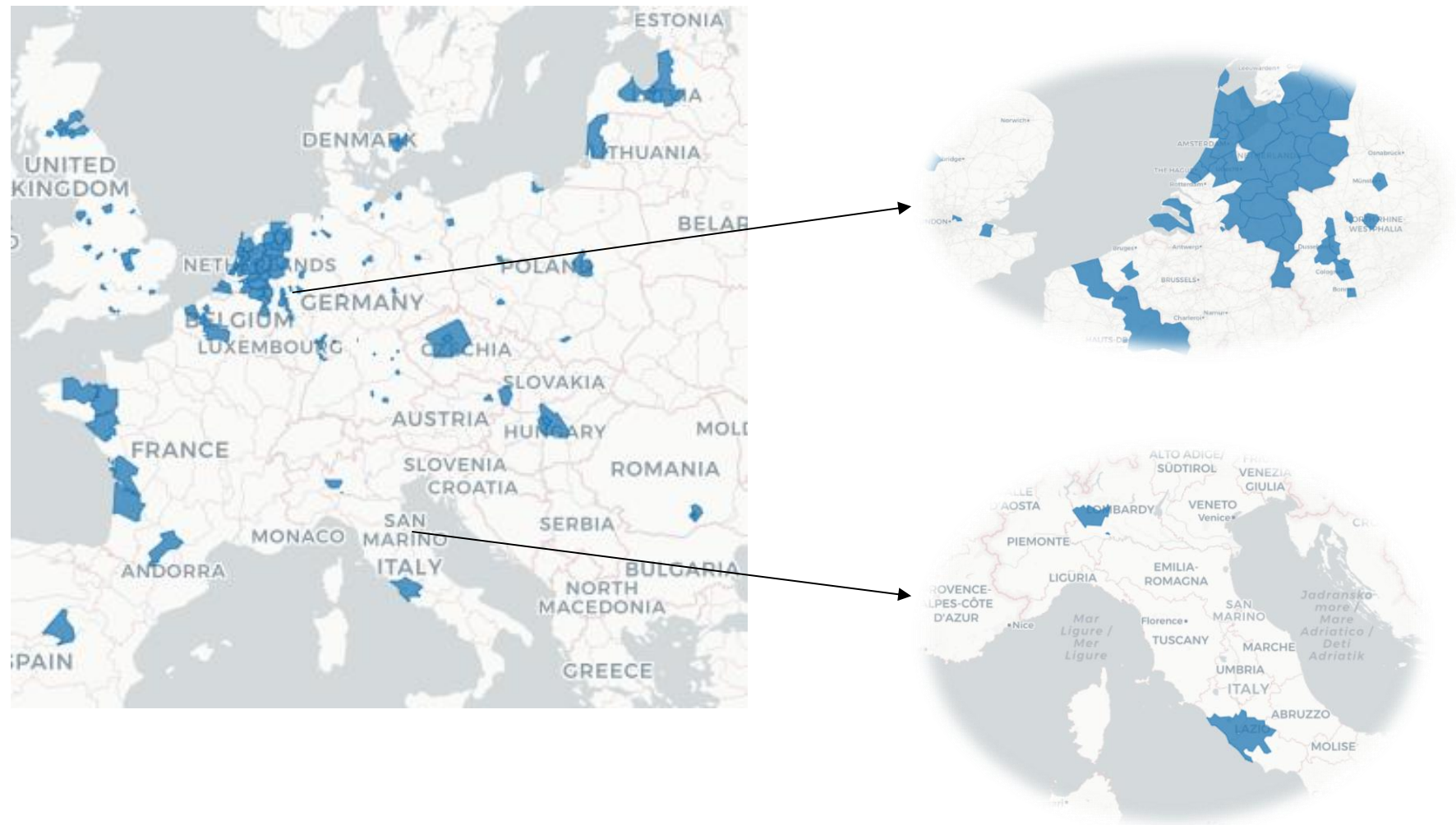
SHAP Summary: Cluster 1
KPCA_RBF_g0p001480|SPEC|Spectral_nn|k10

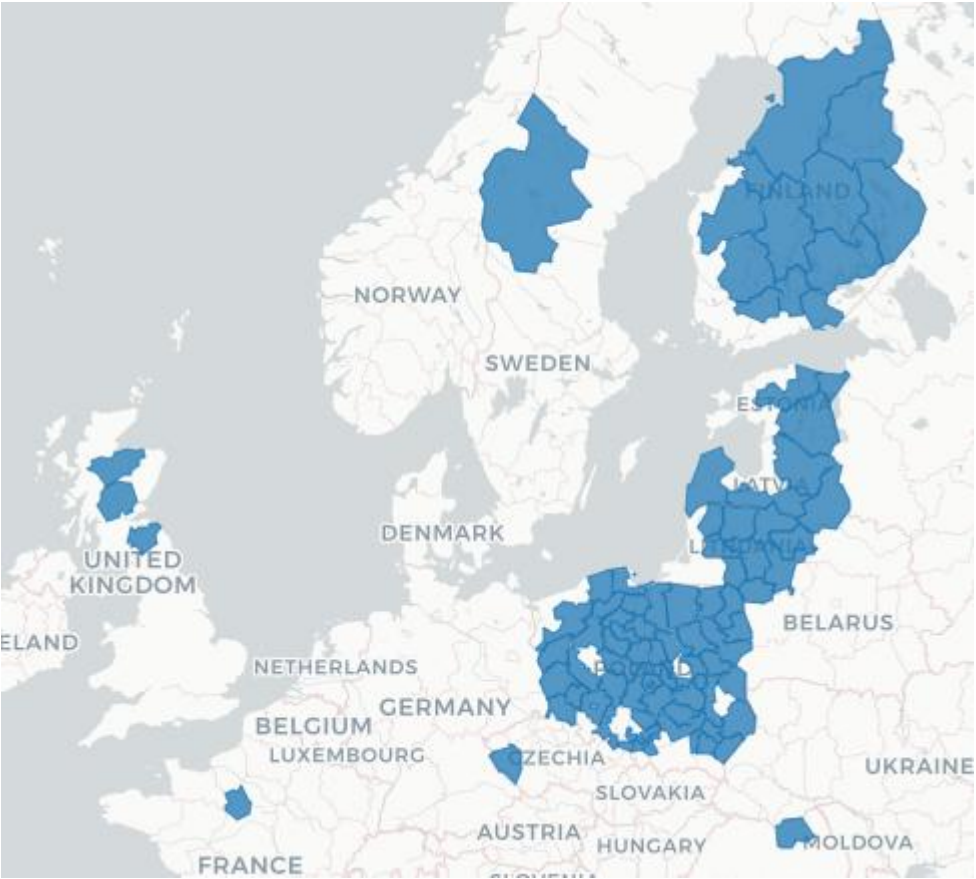




- The population is under stress from rising coastal levels.
- Urban land cover is decreasing.
- There is comparatively less landslide pressure.
- There is an increase in the mean temperature.

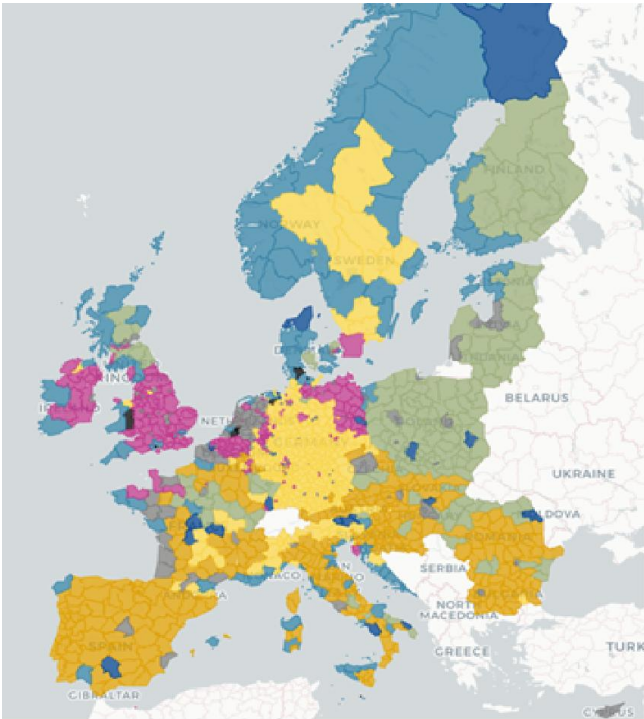
High Urban Land Cover, Metropolitan Areas Under Climate stress





Limited Adaptation and Ageing Population

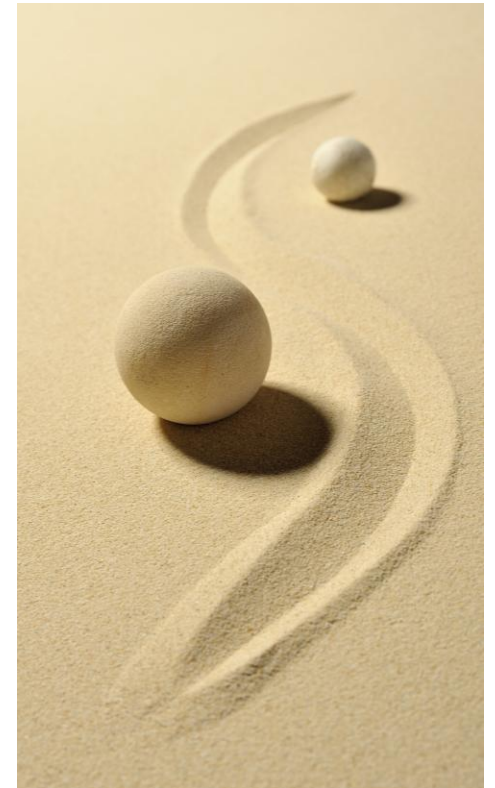
K Means



Cluster	Dominant Drivers (Top SHAP Features)	Theme	Typology Label
0	Lower exposure of settlement population, high urban land cover %, employment share, rail exposure, older population share	Urban Exposure + Socioeconomic Vulnerability	Dense Urban Hotspots (Medium Exposure, Medium Adaptation)
1	High coastal settlement exposure, medium precipitation, rail exposure, employment share	Climate Hazard + Transport Exposure	Climate-Exposed Transport Corridors
2	Change in urban land cover, longer rail length, higher employment share, population exposure to flooding, drought & temperature shifts	High-Activity Industrial Hubs	Economic-Infrastructure Clusters under Urban Transformation
3	Lower urban settlement cover, heat waves, heavy precipitation, extreme weather shifts, rail length, drought hazard	Mixed Climate Pressure	Urban-Industrial Nodes under Compound Hazards
4	Heat and drought zone, urban settlement exposure, airports, elderly share, migration vulnerability	Socio-Demographic Vulnerability + Air Transport Hubs	Ageing Urban Regions with Moderate Exposure
5	Extreme flood history, heat waves, high urbanisation	Compound Flood-Heat Interactions	Climate-Sensitive Coastal/Lowland Agglomerations
6	Landslide-related settlement exposure, water stress, medium airport density	Educated but Landslide- and Water-Stress-Exposed Cores	Developed Nodes under Drought & Hazard Pressure
7	Settlement exposure, very heavy precipitation, airports, young population decline	Dynamic Regions under Extreme Precipitation and Fluvial Risk	Expanding Urban Areas Facing Rainfall Risks

Limitation of Clustering the Entire Dataset

- When applying Dimensionality Reduction (DR) and Clustering together, we lose direct control over which indicators drive the results.
- Some variables dominate naturally, while others though important become less influential in the overall pattern.
- This limits our ability to focus on specific climate hazards or thematic analyses (e.g., flooding, heatwaves).
- Solution: *Move toward scenario-based clustering ,handpicking relevant indicators to study each hazard independently and interpret results more precisely.*





Flooding in Europe: A Scenario Case Study

A depiction of Climate Clustering project

Flooding Indicators Selection



Focus: Indicators that directly reflect fluvial, coastal, and rainfall-driven flooding and their human and infrastructure impacts.

➤ **Hazard Indicators :** capture intensity and frequency of flood events.

Flood return period (RP100), storm surge, consecutive wet days, heavy precipitation days, extreme rainfall deltas

➤ **Exposure Indicators :** quantify how much is at risk.

Population and infrastructure exposed to flooding (settlements, roads, rail, transport nodes)

➤ **Vulnerability Indicators :** adaptive capacity and social resilience.

Migration, children and elderly population ratios, employment share, urban green cover, ports per head — reflect

Flooding Scenario: Results

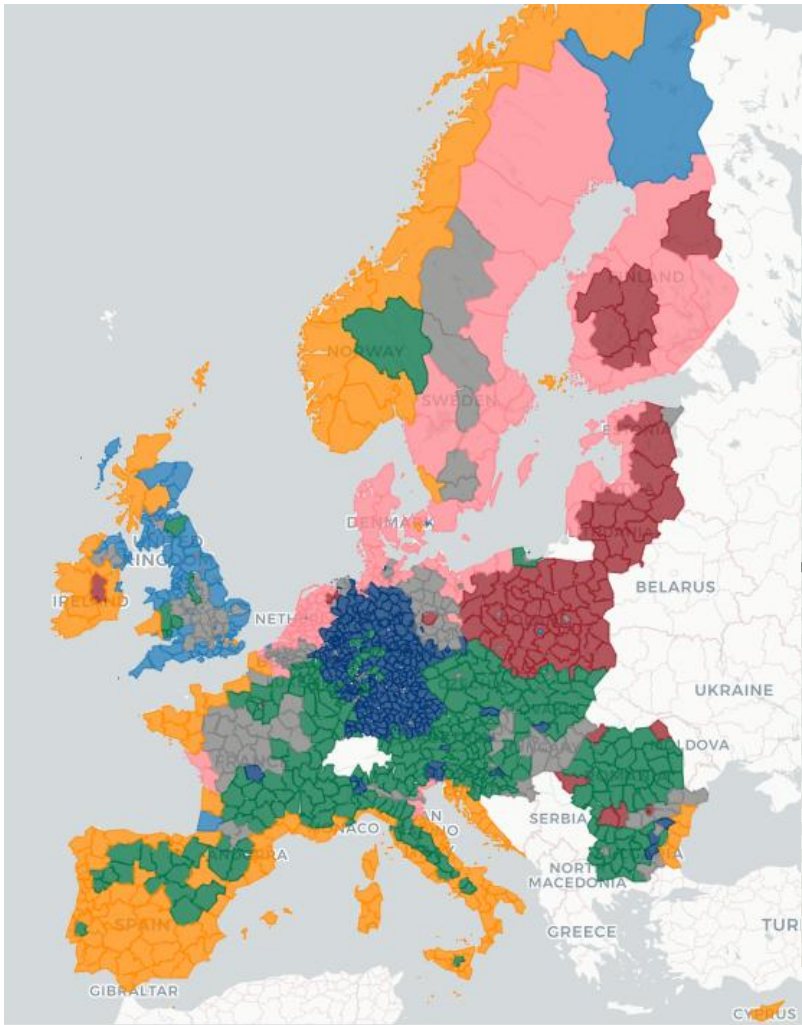
Rank 	Reduction Technique	Clustering Algorithm	Number of Clusters (k)	Silhouette Score	Davies–Bouldin Index	Calinski–Harabasz Index
1	UMAP10	K-Means	7	0.457	0.766	1076.47
2	UMAP10	K-Means	6	0.457	0.793	1062.38
3	UMAP10	Spectral (Nearest Neighbors)	7	0.443	0.772	991.51
4	UMAP10	K-Means	8	0.427	0.834	1085.19
5	UMAP10	Agglomerative (Ward Linkage)	6	0.437	0.773	948.16
6	UMAP10	Agglomerative (Ward Linkage)	7	0.427	0.790	930.84
7	TSNE3	K-Means	8	0.334	0.944	786.73
8	KPCA (RBF Kernel)	K-Means	6	0.159	1.932	128.75
9	PCA	K-Means	8	0.121	1.927	124.49
10	Factor Analysis	K-Means	8	0.123	1.886	106.75

Are our typologies method-consistent?

Similarity Across Clustering Methods ? Are Our Typologies Stable?

Pair	Adjusted Rand Index (ARI)	Interpretation
UMAP10 K-Means k6 ↔ k7	0.89	Very high agreement
UMAP10 K-Means k7 ↔ Agglomerative Ward k7	0.75	Strong match
UMAP10 K-Means k7 ↔ Spectral nn k7	0.82	Very high consistency
UMAP10 K-Means k7 ↔ K-Means k8	0.86	Increasing k adds sub-clusters, not new regimes

After carefully examining each cluster, we can produce our own typology



	Cluster	Typology Name	Core SHAP Drivers	Environmental Character
	0	Humid Low-Exposure Coastal/Port	Low ports, high green, low flood	Stable humid transition zones
	1	Urban Floodplain Expansion	Growing young pop, high urban, low green	Expanding floodplain metros
	2	Industrial & Transport Exposure Corridors	High road/settlement exposure, economic hubs	River/industrial corridors
	3	Urban-Infrastructure Climate Pressure	High rail/road exposure, 1 precipitation	Dense metros under pressure
	4	Ageing Low-Exposure Hinterlands	Ageing pop, low exposure	Demographically vulnerable rural
	5	Coastal Economic Exposure	Ports, flood intensity, heavy rain 1	Coastal trading hubs
	6	Dry Transitional	Low exposure, 1 dry days	Arid transition regions

Purpose of the Scenario Study

- Integrate new data anytime, flexible and future-proof
- Focus on specific themes e.g., flooding, drought, GDP, or population dynamics
- Customize indicators to match real-world relevance
- Compare how different hazards or socio-economic factors shape vulnerability
- Deliver clearer, more interpretable insights than clustering everything together



Any Other scenarios?

Suggestions for Future Scope:

- Implement a comprehensive dashboard/webapp
- Extend framework towards holistic approach going beyond climate adding other risk factors
- Integrate financials and other economical options



Final Conclusion

- Achieved objective of identifying similar European regions using environmental and socio-economic data.
- Recreated and extended the RESIN typology through a fully data-driven approach.
- Validated best-performing clustering combinations with explainability support.
- Introduced scenario-based studies (e.g., flooding) for deeper insights.
- Suggested Future work: Include new scenarios, a Dashboard/ WebApp for direct access.

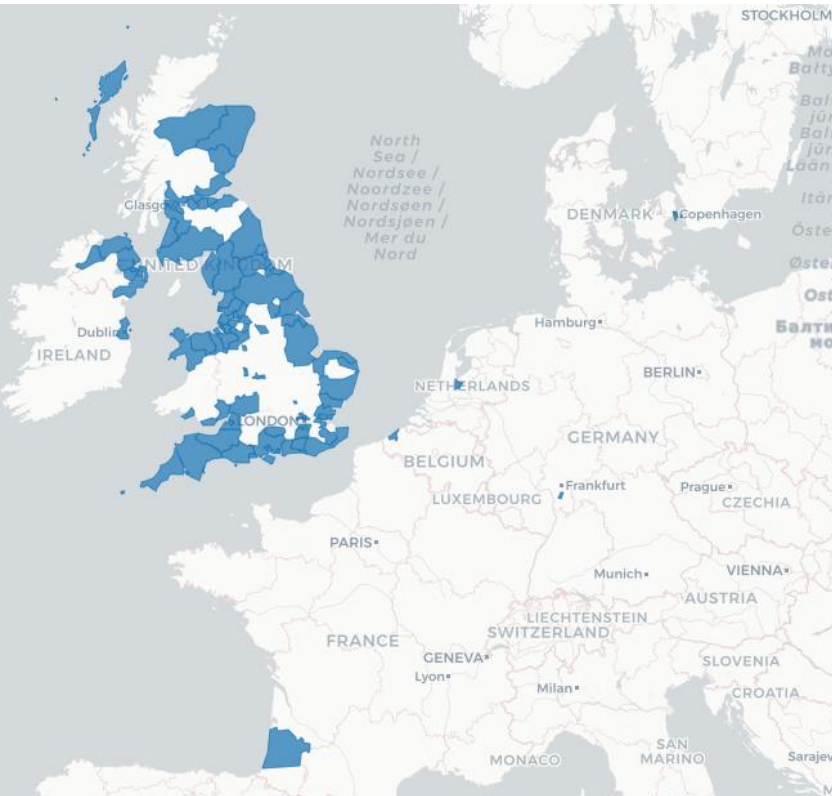


Thank you



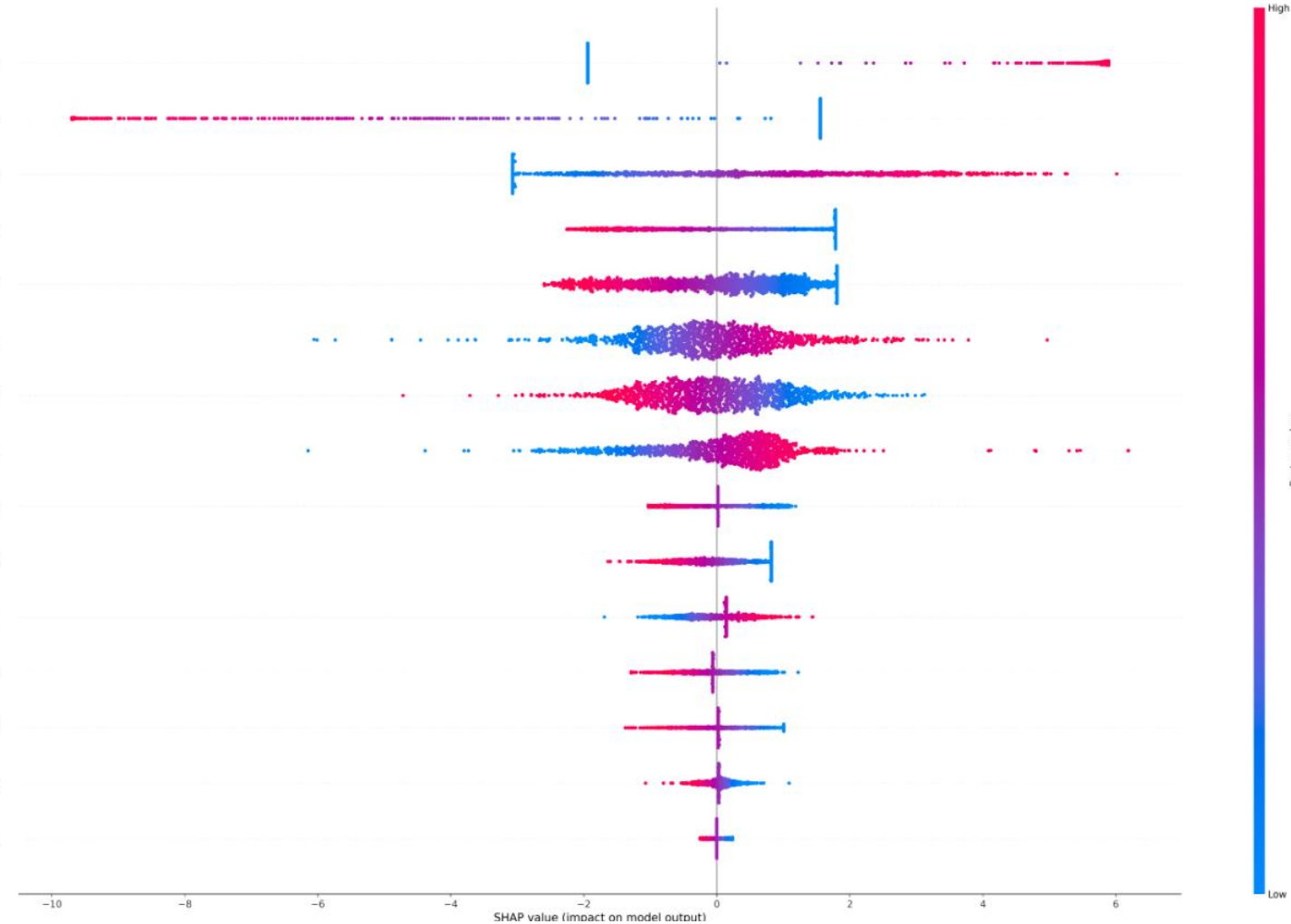
Appendix Sektion:

UMAP (D = 10) + K mean K 7

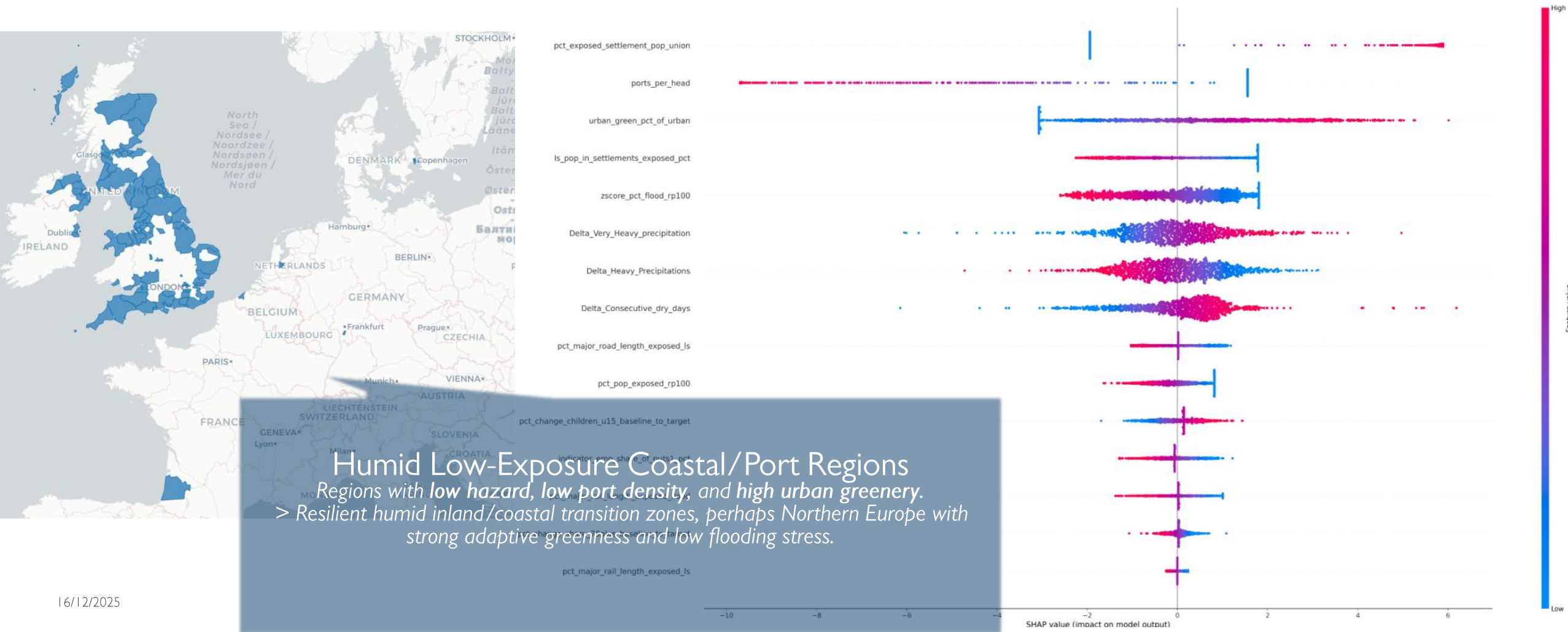


pct_exposed_settlement_pop_union
ports_per_head
urban_green_pct_of_urban
ls_pop_in_settlements_exposed_pct
zscore_pct_flood_rp100
Delta_Very_Heavy_precipitation
Delta_Heavy_Precipitations
Delta_Consecutive_dry_days
pct_major_road_length_exposed_ls
pct_pop_exposed_rp100
pct_change_children_u15_baseline_to_target
indicator_emp_share_of_nuts1_pct
pct_major_rail_length_exposed_flood
pct_change_share_70plus_baseline_to_target
pct_major_rail_length_exposed_ls

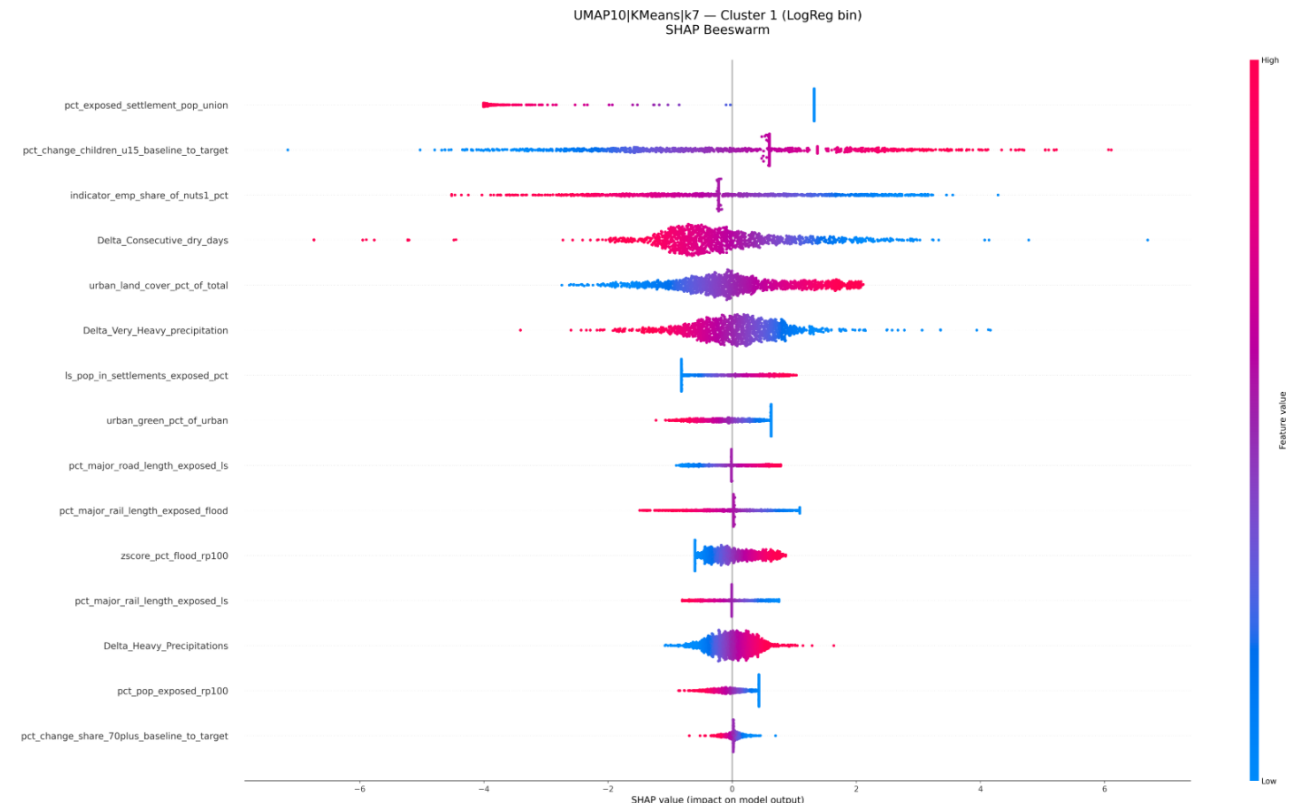
UMAP10|KMeans|k7 — Cluster 0 (LogReg bin)
SHAP Beeswarm



UMAP (D = 10) + K mean K 7



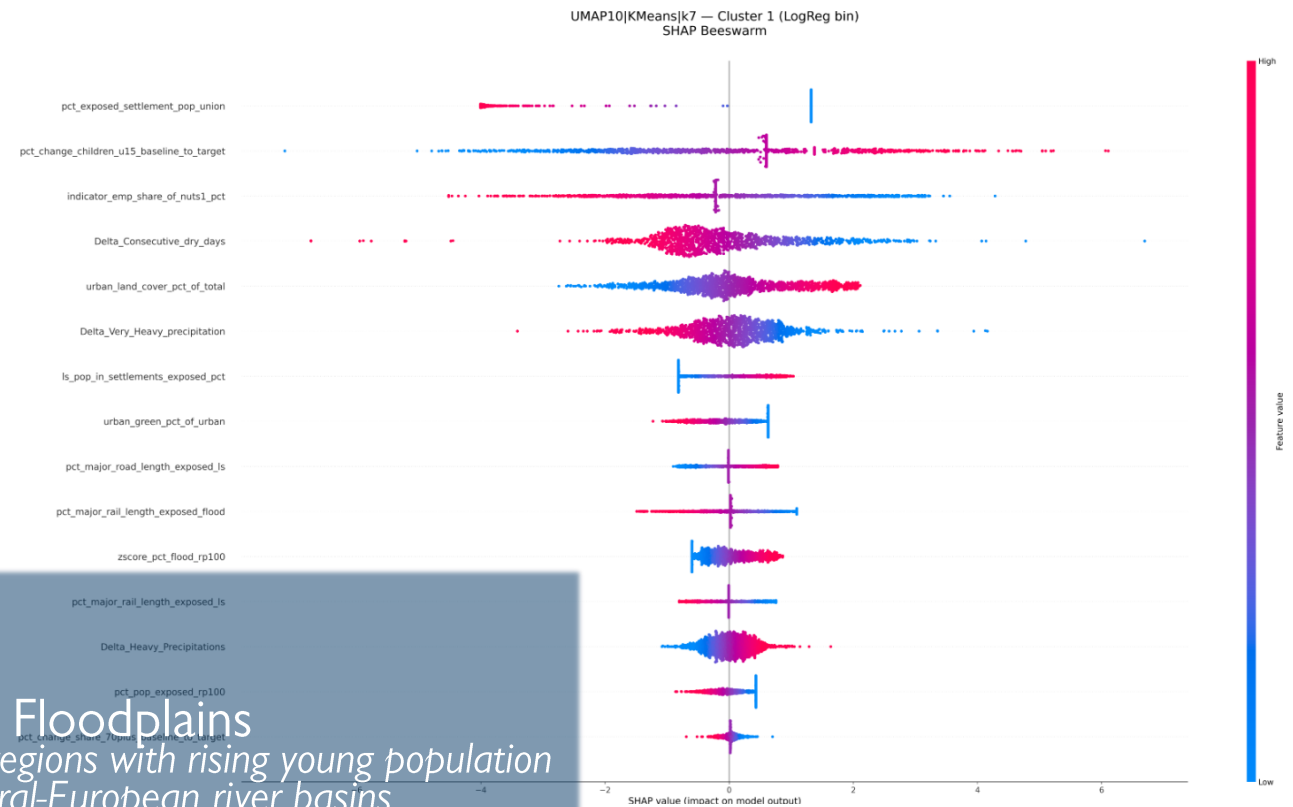
UMAP (D = 10) + K mean K 7



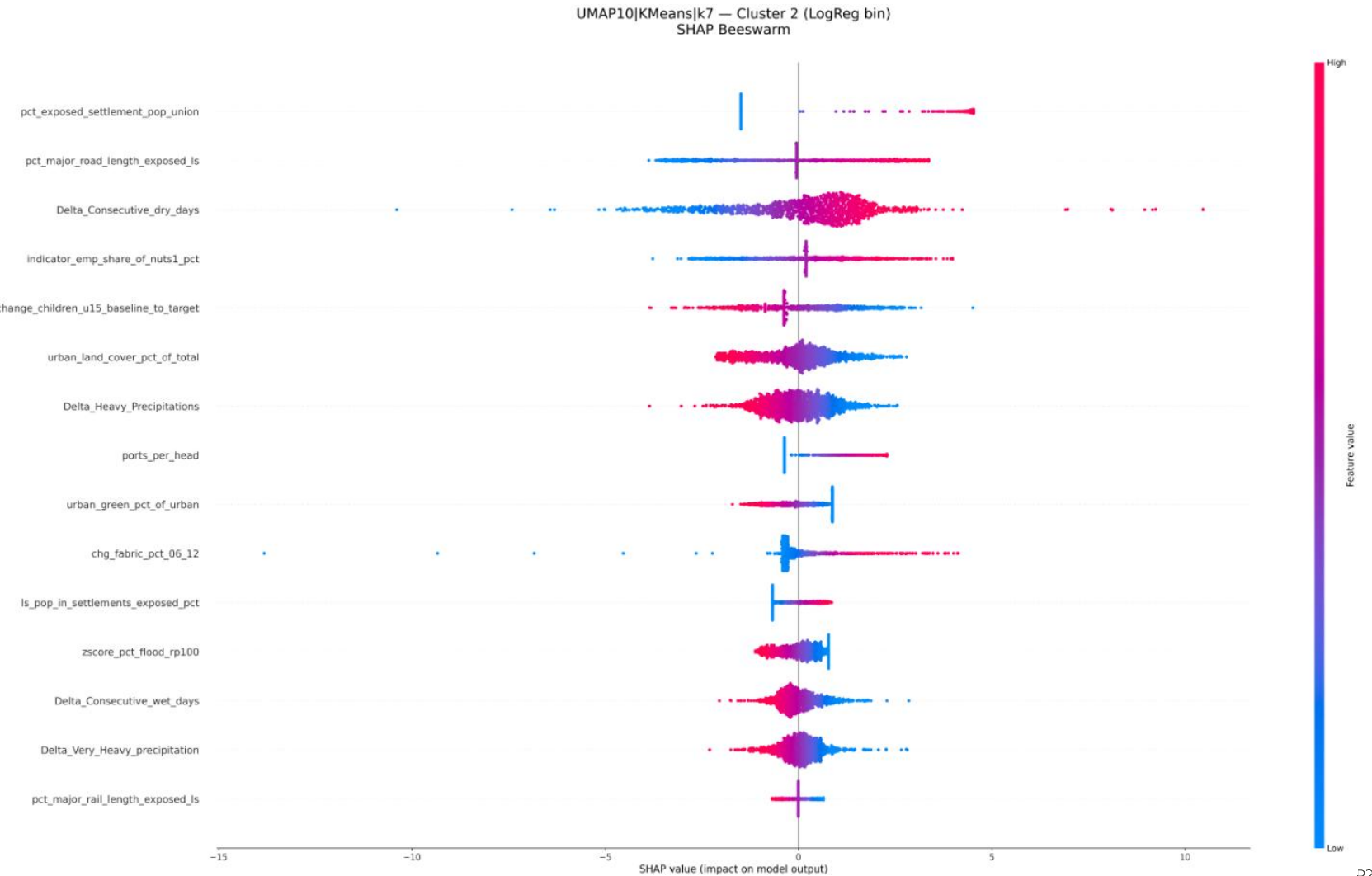
UMAP (D = 10) + K mean K 7



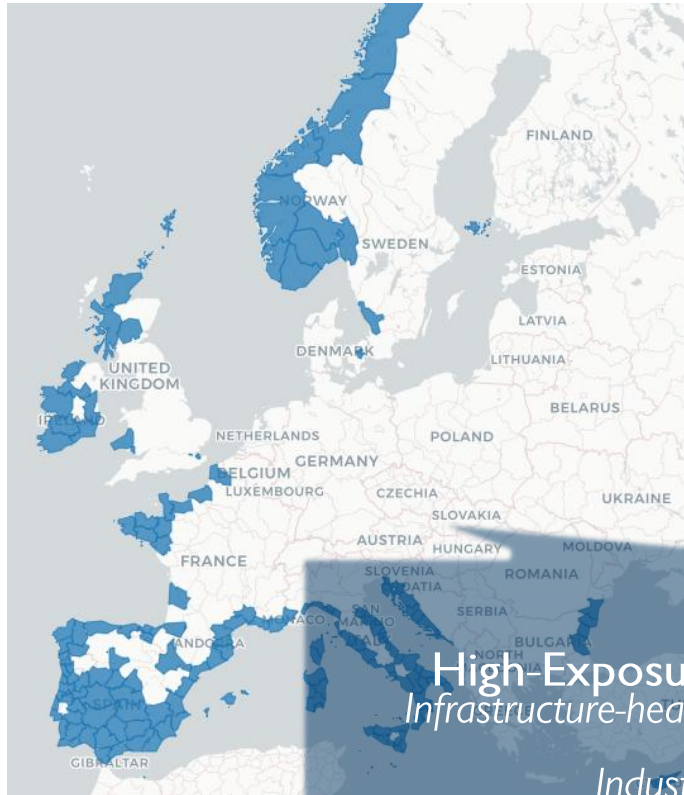
Growing Urban Floodplains
Fast-growing, densely urban floodplain regions with rising young population and limited green cover, Central-European river basins



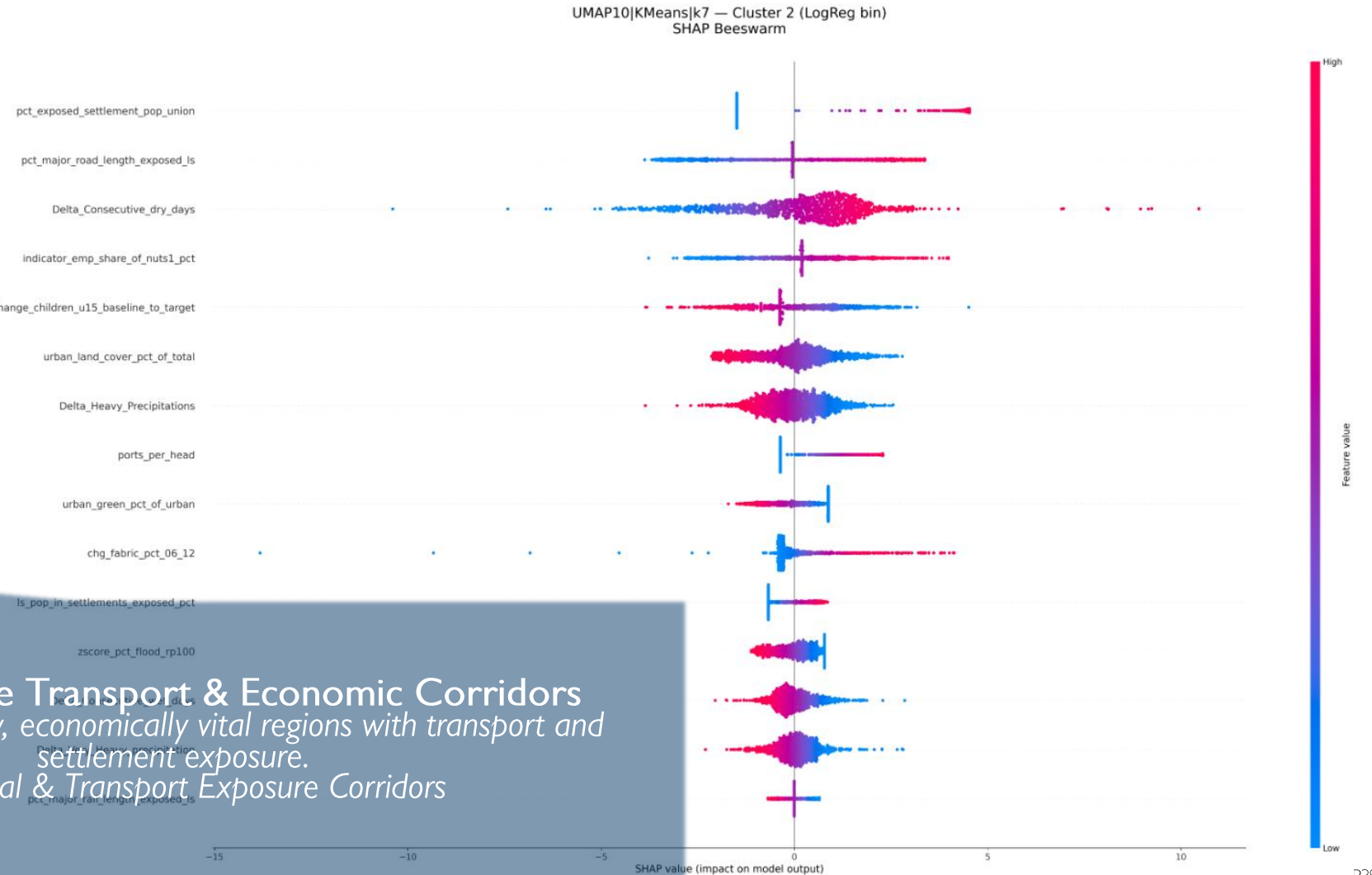
UMAP (D = 10) + K mean K 7



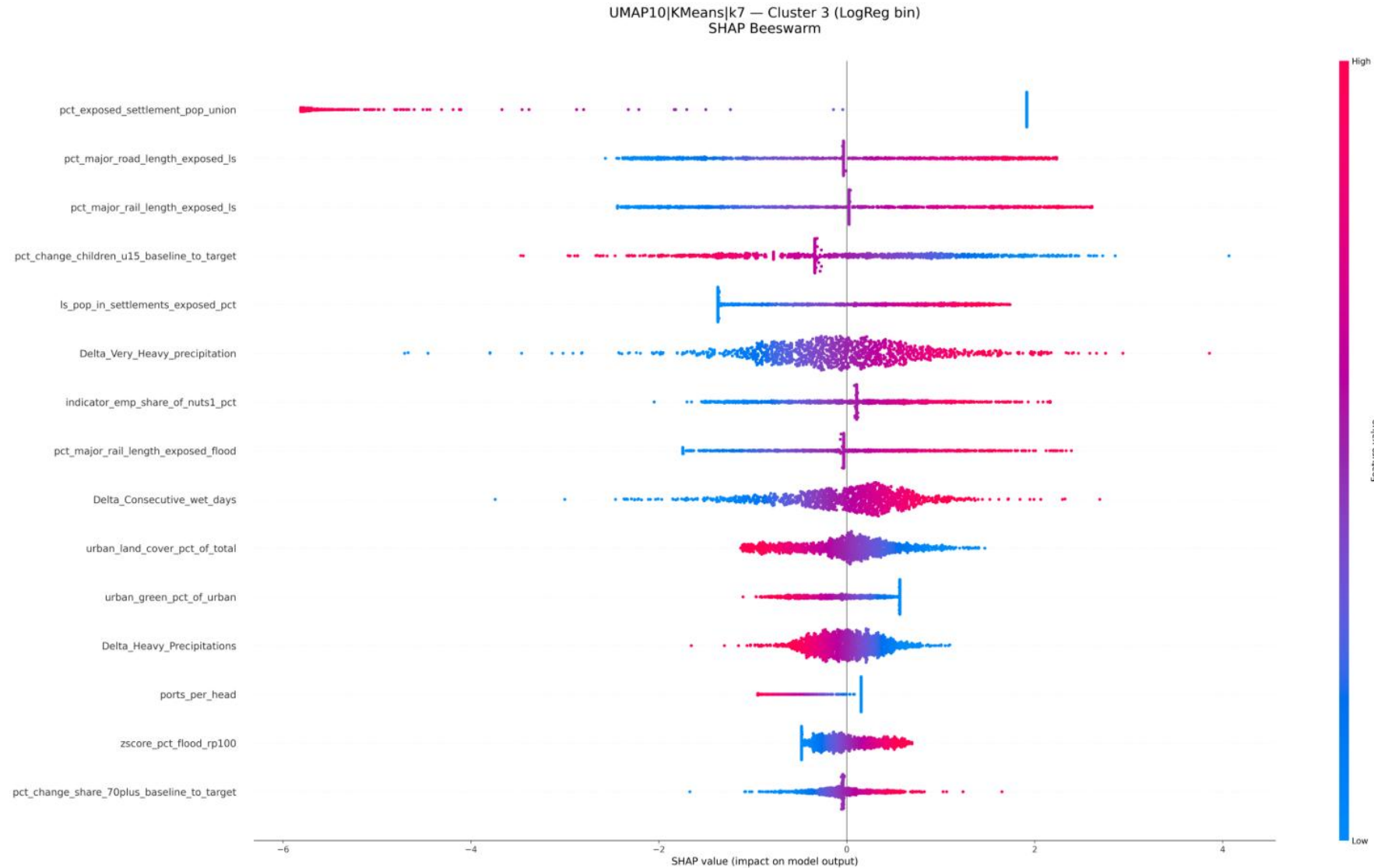
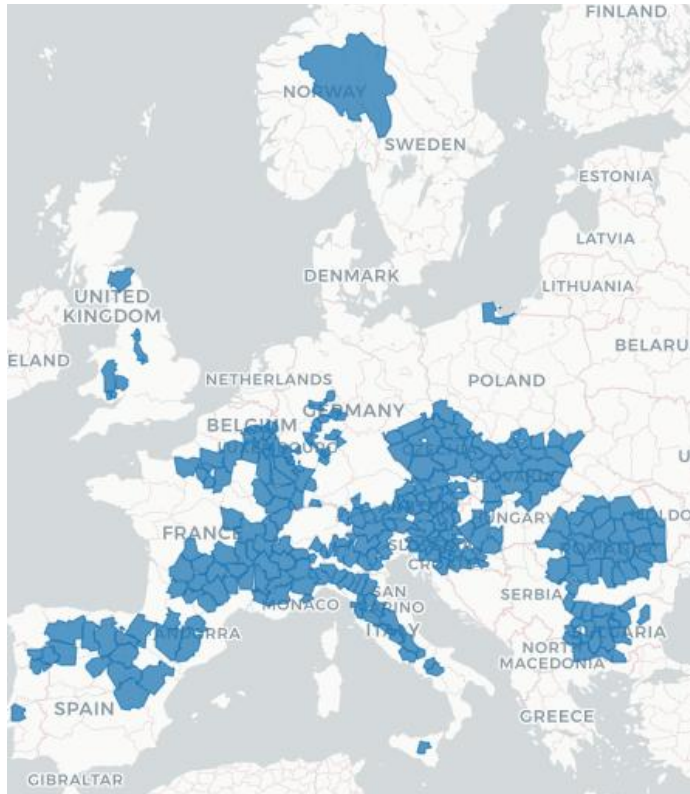
UMAP (D = 10) + K mean K 7



High-Exposure Transport & Economic Corridors
Infrastructure-heavy, economically vital regions with transport and settlement exposure.
Industrial & Transport Exposure Corridors



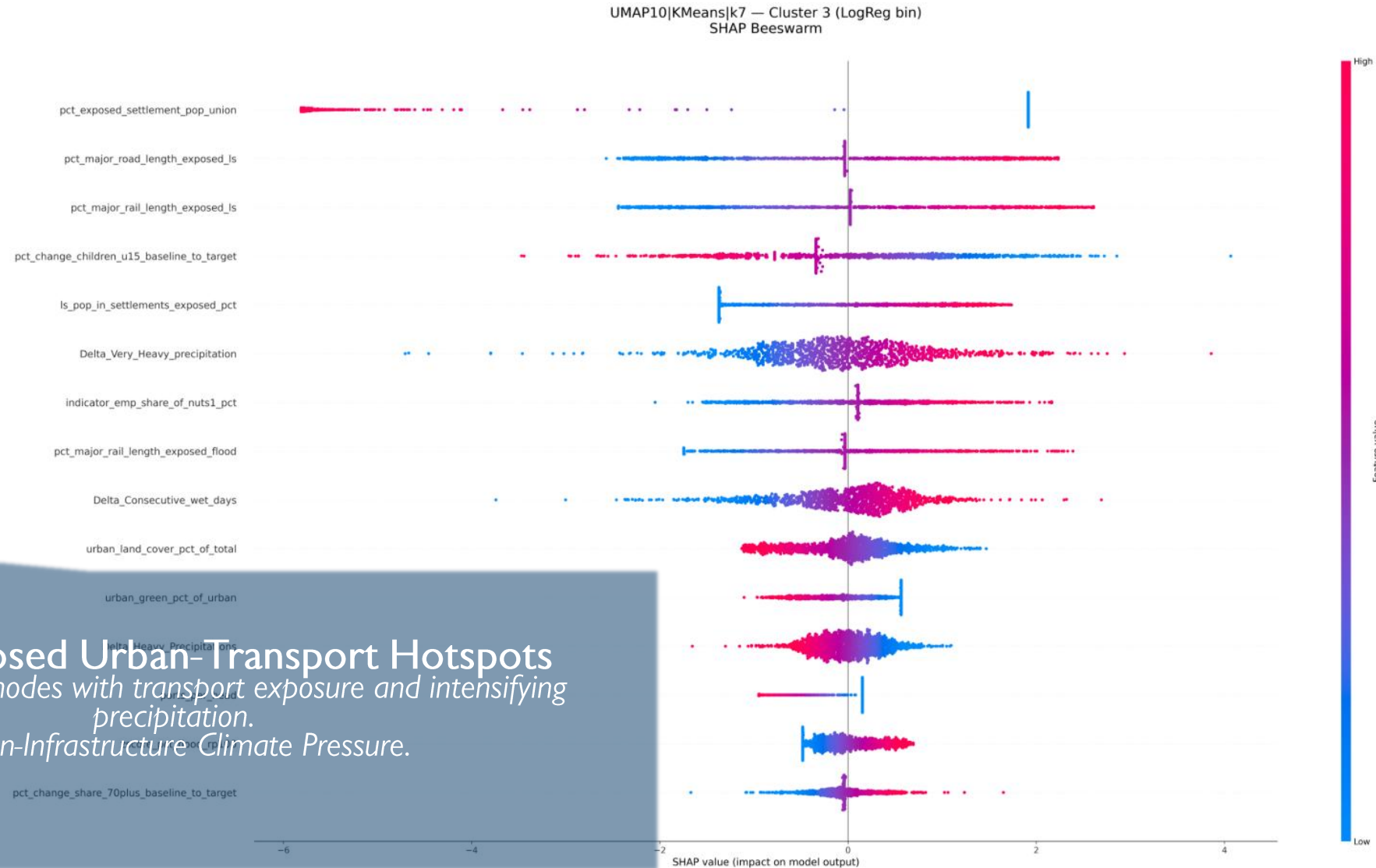
UMAP (D = 10) + K mean K 7



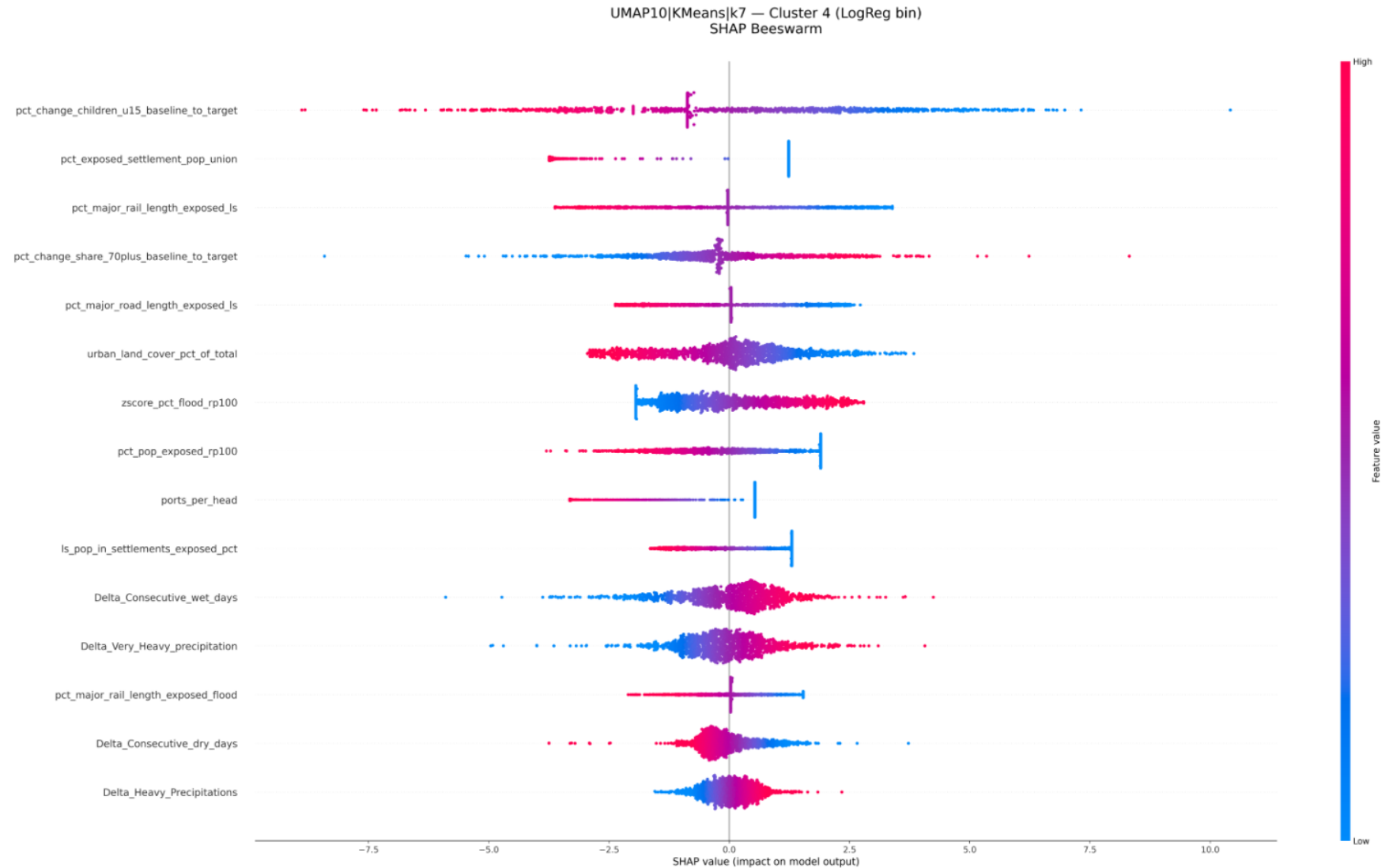
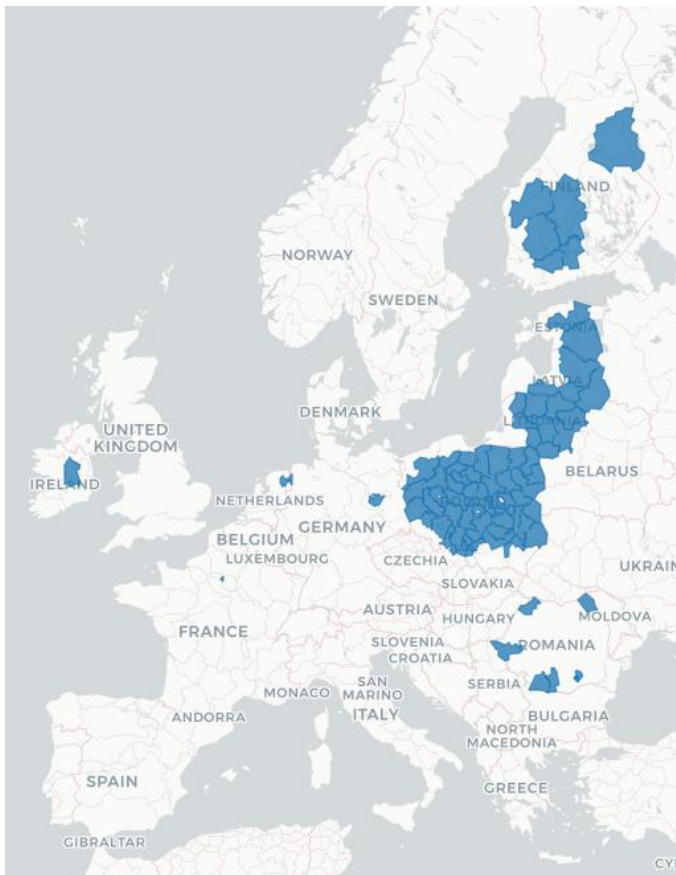
UMAP (D = 10) + K mean K 7



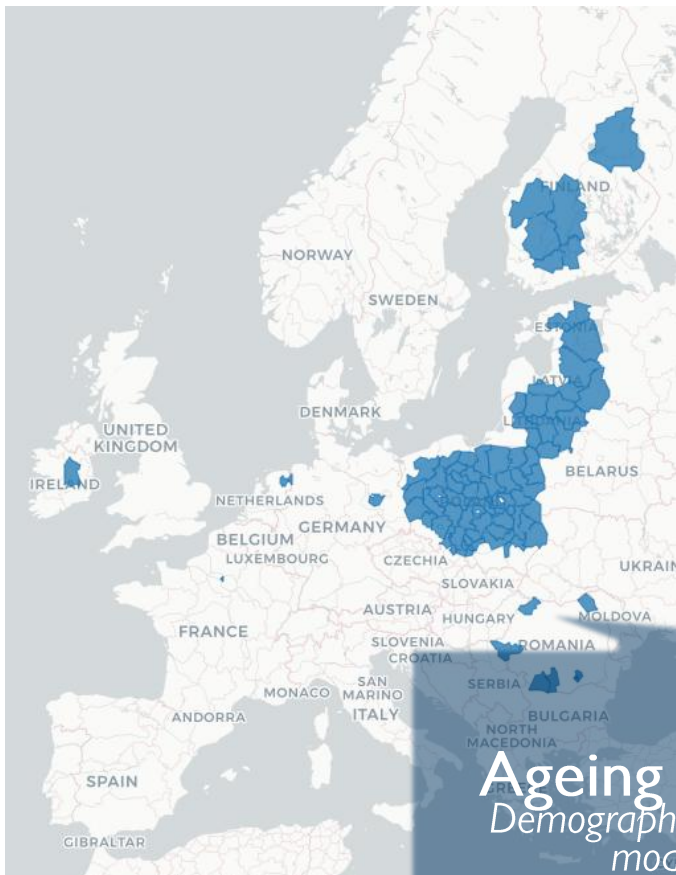
Highly Exposed Urban-Transport Hotspots
Urban-industrial nodes with transport exposure and intensifying precipitation.
Urban-Infrastructure Climate Pressure.



UMAP (D = 10) + K mean K 7



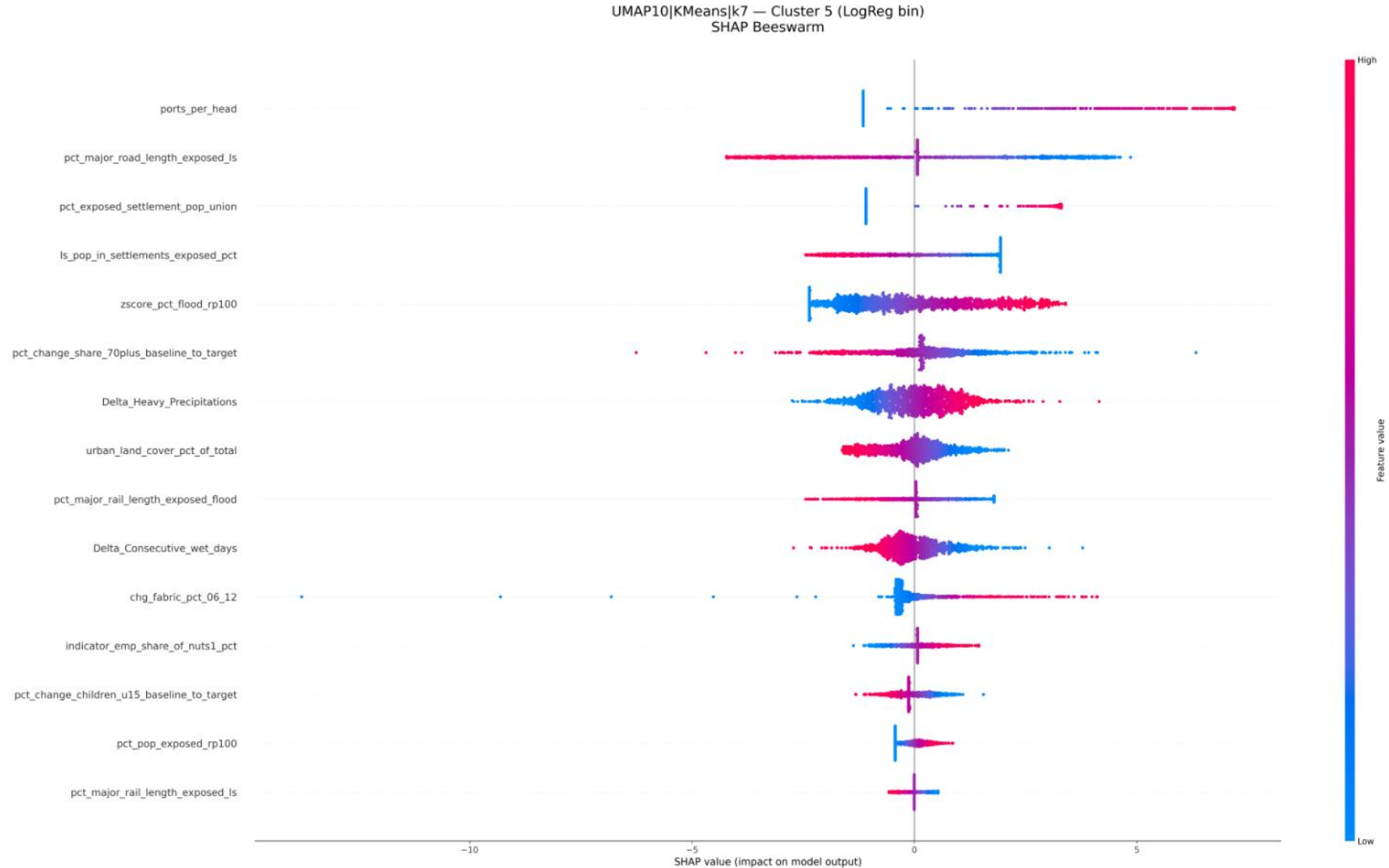
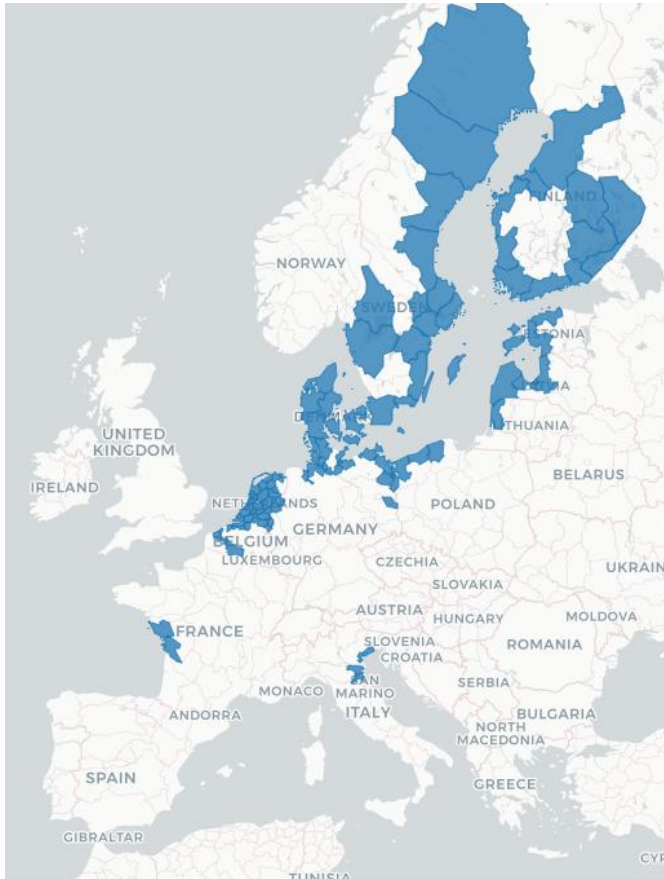
UMAP (D = 10) + K mean K 7



Ageing Low-Exposure Agricultural Inland
Demographically ageing, inland agricultural or peri-rural regions with moderate hazard but limited infrastructure exposure.
Ageing Low-Exposure Hinterlands



UMAP (D = 10) + K mean K 7

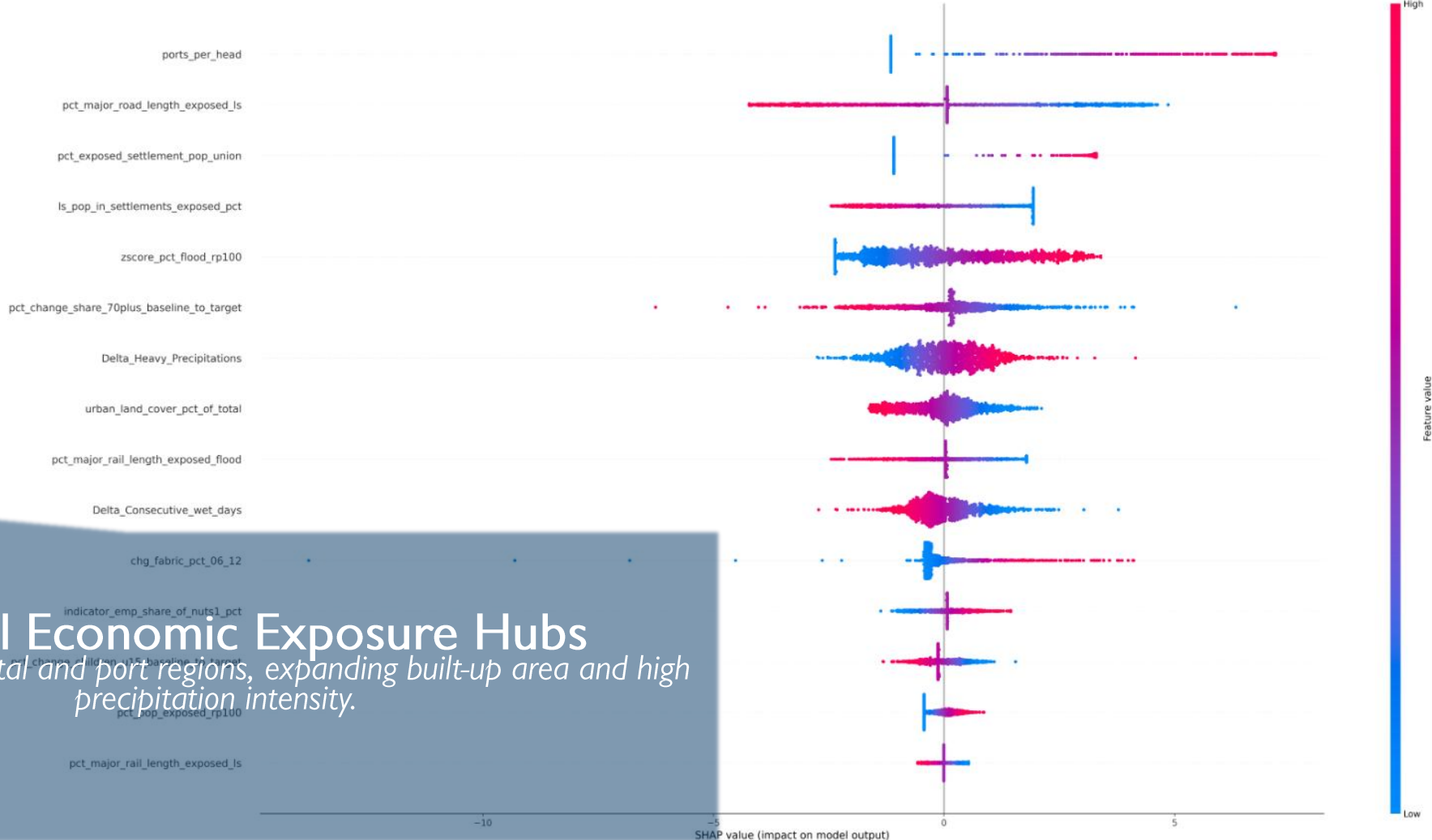


UMAP (D = 10) + K mean K 7

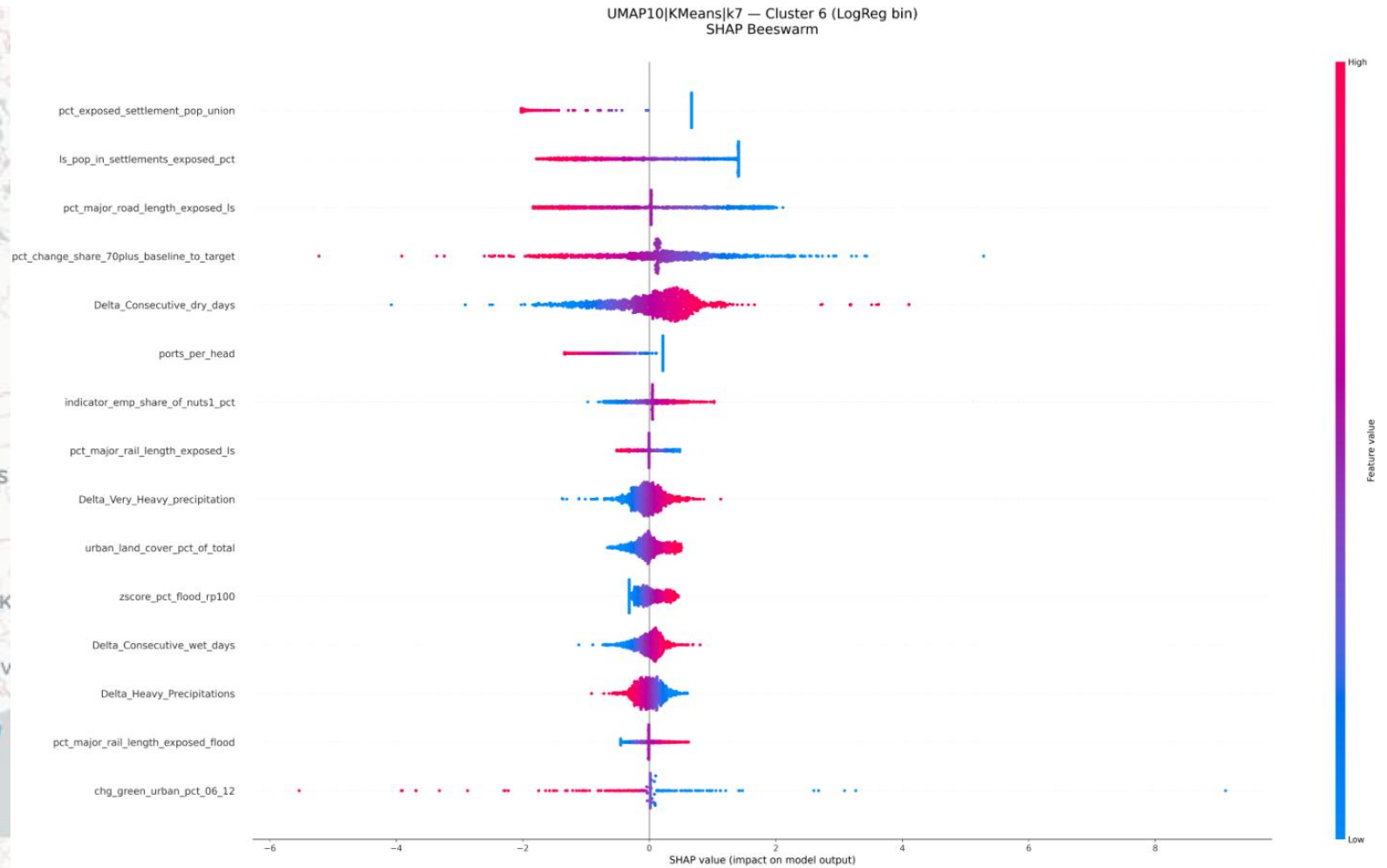
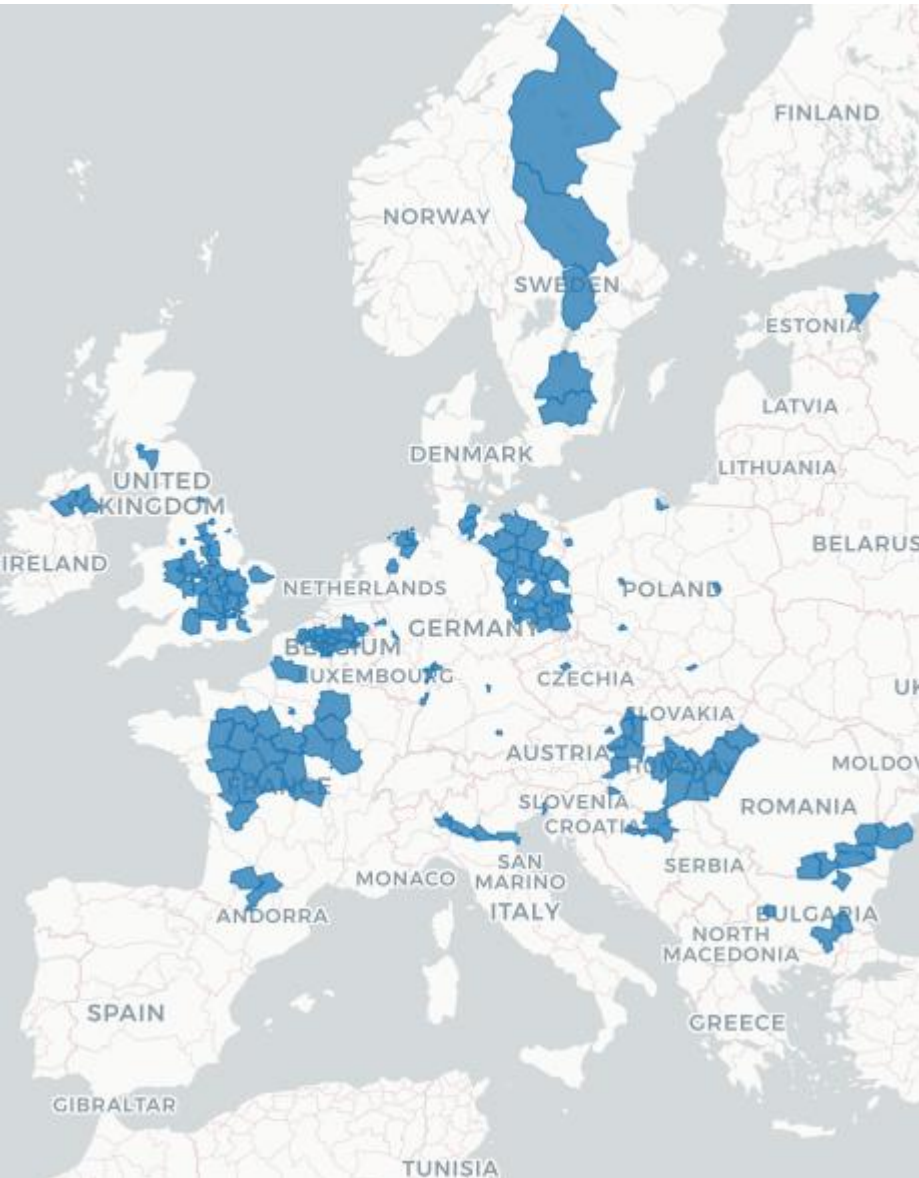


Coastal Economic Exposure Hubs
 High-exposure coastal and port regions, expanding built-up area and high precipitation intensity.

UMAP10|KMeans|k7 — Cluster 5 (LogReg bin)
 SHAP Beeswarm



UMAP (D = 10) + K mean K 7



UMAP (D = 10) + K mean K 7

